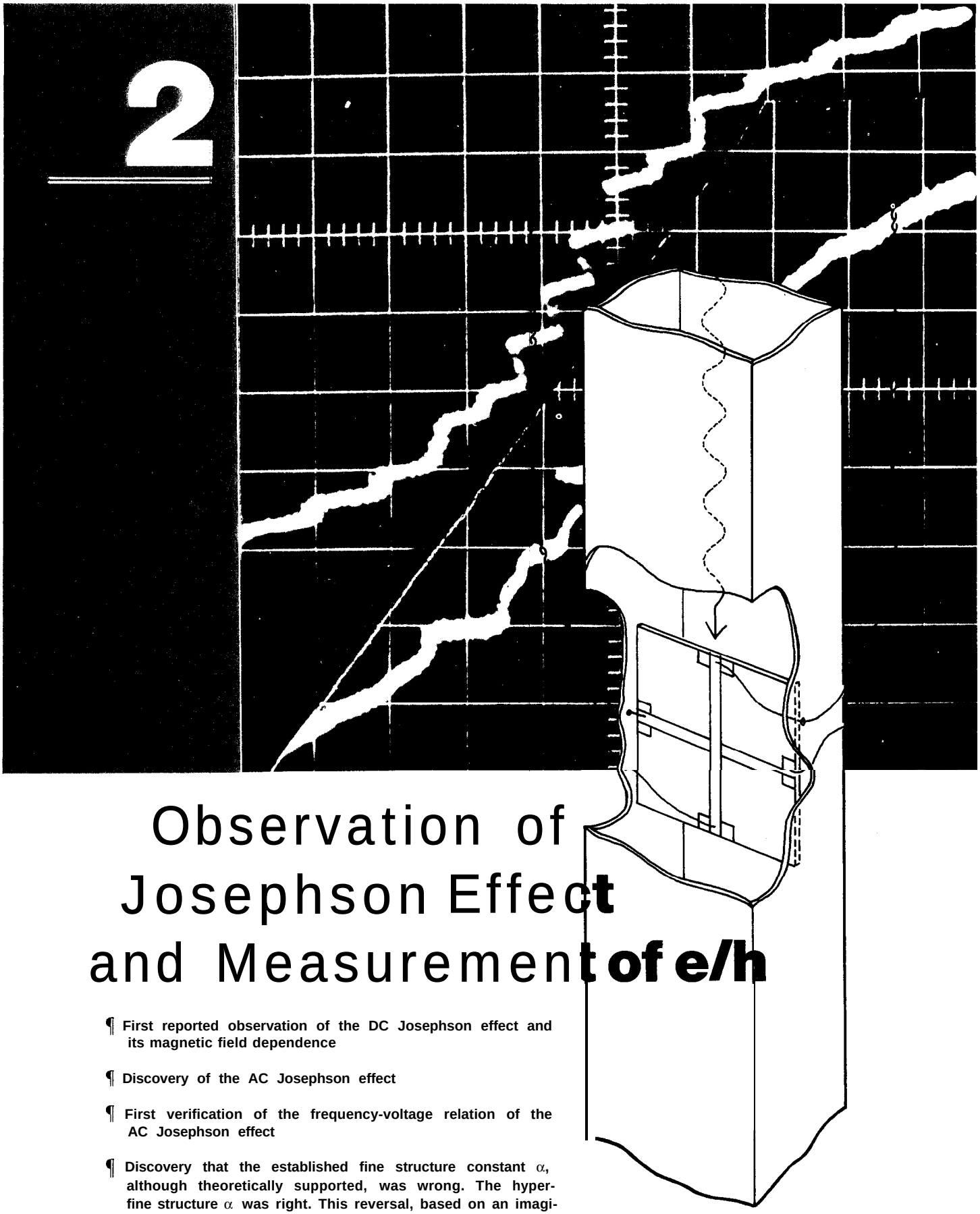
2



Observation of Josephson Effect and Measurement of e/h

- First reported observation of the DC Josephson effect and its magnetic field dependence
- Discovery of the AC Josephson effect
- First verification of the frequency-voltage relation of the AC Josephson effect
- Discovery that the established fine structure constant α, although theoretically supported, was wrong. The hyperfine structure α was right. This reversal, based on an imaginative application of the Josephson effect, removed the discrepancy between the theoretical and experimental values of the hyperfine splitting in the ground state of hydrogen -one of the major unsolved problems of quantum electrodynamics at that time (1966).

A NOTE ON THE NOBEL PRIZE FOR TUNNELING

Tunnel Effect

The tunnel effect in semiconductors and insulators is a special case of the general quantum-mechanical phenomenon of the passage of particles through potential barriers. Classically, a particle can never cross a region of space where there is a potential barrier whose height V_o exceeds the particle's total energy E, because its kinetic energy inside the barrier would be negative. However, when the wave aspect of the particle is taken into consideration, there exist, in the barrier, solutions of Schrodinger's equation which decay exponentially with depth of penetration. The particle has thus a finite probability of being found on the far side of the barrier, as if it had 'tunneled' through the potential mountain.

Probably the best known case of tunnel effect is the passage of α-particles through the potential barrier surrounding atomic nuclei. But the effect occurs very widely: for instance, in nuclear reactions, in many molecular phenomena, in field emissions from metals, etc.

In semiconductors and insulators, one means by 'tunnel effect' the internal field emission first considered by Zener in 1934 to explain dielectric breakdown in dielectrics (though it is now known that ordinary breakdown occurs generally through different mechanisms). For an electron in an insulator, the gap between the conduction band and the valence band behaves as a potential barrier. The width of the barrier is determined by the magnitude of the field. There is thus a finite probability for the electron, without changing its energy, to tunnel across the forbidden gap into the empty conduction band. This process creates an electron-hole pair.

Based on Encyclopaedic Dictionary of Physics (Pergamon Press, 1962)

Josephson Tunneling *versus* Single Particle Tunneling

As this chapter was being prepared for publication, the 1973 Nobel Prize for Physics was announced. The award is relevant to the subject of this chapter, as it was shared by three physicists for their work on tunneling effects. One half of the prize went to Brian Josephson for his theory of 'Josephson Tunneling,' which is the subject of this chapter; the other half was shared by Leo Esaki and Ivar Giaever for their earlier experimental work on single particle tunneling (Esaki in semiconductors and Giaever in superconductors).

Josephson tunneling is the tunneling of *correlated* electron pairs ('Cooper' pairs), while 'ordinary' or single particle tunneling is the passage of *uncorrelated* single electrons through *insulator* barriers. (There is also the tunneling of both kinds through metal barriers, but its mechanism is quite different in detail and not within the scope of this chapter.)

While Josephson tunneling can occur only between two superconductors (which must be at very low temperatures), single electron tunneling can take place at any temperature, provided the barrier is thin enough. Since, in the latter, each electron is, not correlated with any other electron, the system manifests a normal resistance as opposed to Josephson tunneling where there is zero resistance.

There is also a quantitative difference in the insulator barrier thickness between the two types of tunneling: Josephson tunneling requires much thinner barriers, typically 10Å, whereas single particle tunneling occurs through thicknesses of several hundred Å.

There are some interesting parallels as well as contrasts in the way in which the two effects were discovered.

Prior to the experiments of Giaever, many theorists did not believe the superconducting energy gap to be observable by means of single particle tunneling. The observation was made experimentally by a physicist who did not have confidence in the theoretical predictions, and indeed proved these predictions to be incorrect.

Josephson predicted the Cooper pair tunneling and worked out its complete theory while a graduate student in *experimental* physics in Cambridge, England; yet, his attempts to verify it in the laboratory were not successful at the time, and thus it was not believed. As the reader will see in this chapter, the effect in its various forms was experimentally demonstrated in the U.S.A., after one of its few believers-theorist Anderson-brought and communicated the ideas of Josephson to experimentalist Rowell, (see "How Josephson's Idea Took Hold in America," p. 49).

EXPLANATORY PHYSICS NOTES FOR NON-SPECIALISTS

Josephson Effect

In April-June 1914, H. Kamerlingh-Onnes, the discoverer of superconductivity (1911), electromagnetically induced a current in a loop of superconducting wire. He found that the current flowed continuously for long periods without decay. In fact, this current persisted so long that he was able to transport a superconducting loop of current from Leiden to Cambridge.

Had Kamerlingh-Onnes replaced a section of his resistanceless superconducting wire with a thick section of dielectric, the current would have ceased, because the dielectric has resistance. With a thin enough section of dielectric (of the order of several tens of Angstroms), he would have been able, however, to observe what is now known as the Josephson effect.

In 1962, Josephson predicted[1] that paired electrons could tunnel through a thin dielectric barrier between two separate superconductors. This 'Josephson effect' is purely a quantum-mechanical phenomenon and has no classical analogue. In this phenomenon, the *phase* of the wave function is of decisive importance, contrary to the many examples to which one is often introduced in elementary quantum mechanics. The Josephson effect demonstrates the coherence of the quantum phase that can, under the proper conditions, be established between weakly coupled superconductors.

1. Josephson, B.D., *Phys. Letters* 1, 251(1962).

One way in which two superconductors may be weakly coupled is by inserting an insulating barrier (through which superconducting electrons may tunnel) between them. The barrier must be sufficiently thin so that the superconductors are in close enough proximity to be coupled. They must not be too close to one another (within several Angstroms), for then they will be strongly coupled. The Josephson effect occurs only in the case of weak coupling. Since electron *pairs* are the basis of superconductivity, the coupling is achieved by the *tunneling of* a superconducting electron pair through that barrier.

Two Forms of the Josephson Effect: DC and AC

This paired electron tunneling current has been shown to have two forms:

DC: A direct supercurrent (up to some maximum value) can flow without the existence of any external voltage across the barrier. The zero-resistance property of a superconducting system has a physical meaning as a quantum-mechanical phase phenomenon, having to do

with electron pairing. This phase difference across the barrier can give rise to a current with no applied voltage difference across the junction. This current is described by Josephson's equations (see below).

Some time before Josephson's prediction, physicists working with superconducting tunnel junctions were frequently observing annoying 'shorts' which, we now know, may have been the Josephson current at zero applied voltage. An example of this type of phenomenon was published by Nicol, Shapiro and Smith[2] in 1960, but since Josephson's theory did not exist at this time, the observations were explained as 'a characteristic exhibit of a negative resistance region.' (See Fig. 1 of Discovery Story 1 by Shapiro in this chapter.) On the other hand, real 'shorts' can exist due to pinholes in the insulating layer. These pose an experimental problem to those physicists now studying the Josephson effect. Thus, while the effect of Nicol, Shapiro and Smith may be interpreted as a Josephson effect, it also has some of the characteristics of a 'short.'

2. Nicol, James; Shapiro, Sidney; and Smith, Paul; *Phys. Rev. Letters* 5, 461 (1960).
3. Anderson, P.W., and Rowell, J.M., *Phys. Rev. Letters* 10, 230 (1963).

The first paper which claimed to observe the DC zero-voltage current was published in 1963 by Anderson and Rowell. This paper[3], "Probable Observation of the Josephson Superconducting Tunneling Effect," is reproduced in this chapter.

AC: For this presentation, the AC Josephson effect can be divided into two types: DC-biased and RF-biased.

DC-Biased: If two weakly coupled superconductors are maintained at a DC potential difference V, there exists an alternating supercurrent across the junction with frequency

$$\nu = 2eV/h$$

Strictly speaking, eV is a 'chemical potential difference.' The equation may also be written as $\nu = e^*V/h$ where e^* is the charge of the superconducting 'particle.' In this case, $e^* = 2e$, since the 'particles' are bound electron pairs.

RF-Biased: If an alternating supercurrent is frequency modulated by an applied rf field, this will lead to a region of current steps in the DC I-V characteristic at bias voltages which are integer multiples of hf/2e where f is the applied radio frequency, namely

$$V_n = nhf/2e \qquad n = 0, 1, 2, \ldots$$

HOW JOSEPHSON'S IDEA TOOK HOLD IN AMERICA

Synopsis

-Philip Anderson was with the solid state theory group in the Mond Laboratory, Cambridge, at the time Brian Josephson (then a graduate student in experimental physics with Brian Pippard his thesis advisor) developed the theory of his effect. Not long after Josephson's paper appeared in the European journal *Physics Letters* (then a new journal), Anderson returned to America. In the following excerpt from an article written by Anderson*, the reader will learn how the ideas of an unknown physicist published in a then unknown journal were transmitted and accepted here:

* **Anderson, P.W., "How Josephson Discovered His Effect," *Physics Today*, November, 1970, p. 23.**

After a few discussions at the Cavendish, which did not carry me much further than what had been published in Josephson's first paper, I returned home to Bell Labs. There I mentioned to Rowell my conviction that Josephson was right. Rowell admitted to me that he had, from time to time, seen suggestive things in his tunnel junctions and a few months later he called me in to look at his experimental results on a new batch of tunnel junctions. He thought he might actually be seeing the Josephson effect. In thinking at that time about the results of Rowell's measurements, the penny finally dropped for me, and the only two remaining essential components of the phenomenon became clear.

The first of these is that $J = J_1 \sin \phi$ implied a coupling energy between the two superconductors of $E = E_1 \cos \phi$. The value of E_1 is a very vital point; it must be larger than the noise energy in the circuit, which is why our experiments with low-resistance junctions worked and previous attempts by Josephson and others with higher resistance junctions, and thus smaller J_1 and E_1, had not been successful. The second new idea was the Josephson penetration depth.

Looking back, I assume that many workers must have observed the coupling-energy effect previously, but the point is that this effect is not easy to distinguish from a tunnel junction shorted across the layer separating the two metals. We were able to see the effect because three conditions were satisfied: first, we knew what to look for; second, we understood what we saw. (Both of these were the result of our contact with Josephson.) The third condition was that we were confident of Rowell's skills in making good, clean, reliable tunnel junctions.

P. W. Anderson

PROBABLE OBSERVATION OF THE JOSEPHSON SUPERCONDUCTING TUNNELING EFFECT

P. W. Anderson and J. M. Rowell
Bell Telephone Laboratories, Murray Hill, New Jersey
(Received 11 January 1963)

Josephson[1] has recently predicted theoretically a new kind of tunneling current through an insulating barrier between two superconducting metals, This anomalous current behaves like a direct tunneling of condensed electron pairs between the Fermi surfaces of the two metals. When the voltage difference across the barrier is zero, the current should be dc but may range between limits of the order of magnitude of the usual tunnel current above the gap, depending on the relative value of the phase of the energy gap function or "second Green's function F" on the two sides, while at a nonzero voltage difference ΔV it is alternating at a frequency $2e\Delta V/h$.

We have observed an anomalous dc tunneling current at or near zero voltage in very thin tin oxide barriers between superconducting Sn and Pb, which we cannot ascribe to superconducting leakage paths across the barrier, and which behaves in several respects as the Josephson current might be expected to.

Figure 1 shows an X - Y recorder plot of the tunneling current vs voltage for one of these structures at ~1.5°K. The lead and tin films are both approximately 2000Å thick, and the junction has dimensions 0.025 x 0.065 cm^2 and a resistance (both metals normal) of 0.4Ω. Voltage is applied to two arms of the junction from a 1-kΩ potentiometer and the resulting current flow is measured as voltage across a series resistor of 10Ω. The voltage appearing across the barrier is taken directly from the other two arms of the junction. Figure 2 shows the plot with current scale expanded to show the anomalous region near the origin. The current

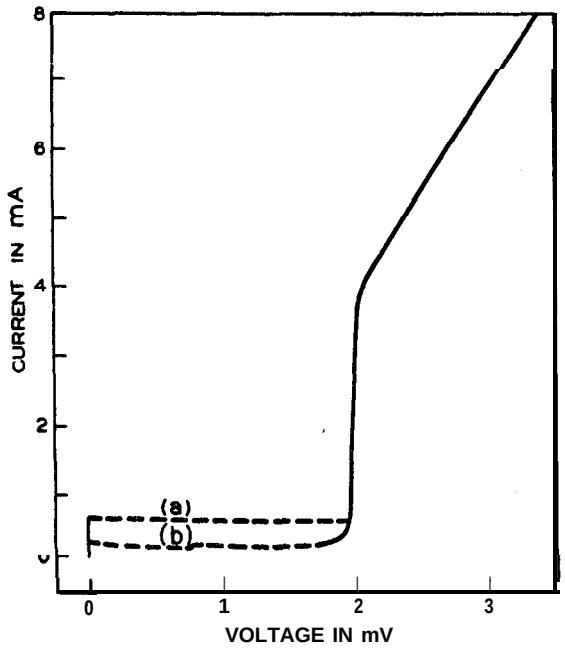


FIG. 1. Current-voltage characteristic for a tin-tin oxide-lead tunnel structure at ~1.5°K, (a) for a field of 6 x 10^{-3} gauss and (b) for a field 0.4 gauss.

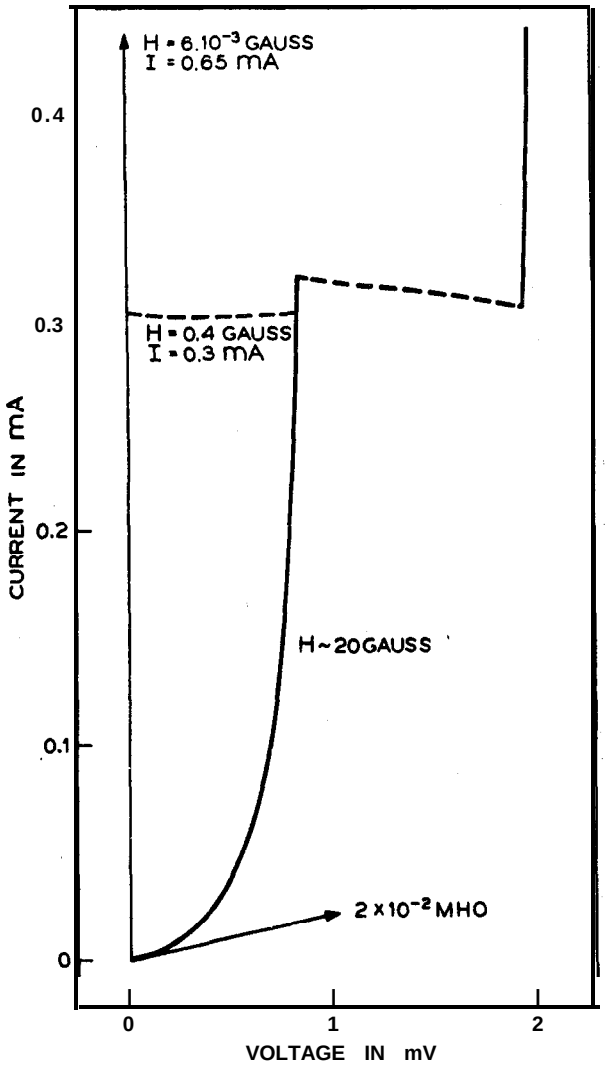


FIG. 2. Current-voltage characteristic for structure of Fig. 1 with expanded current scale. Note the conductance at low voltages in a field ~20 gauss.

at first increases up to a value of 0.3 mA with no voltage appearing across the barrier. At this point the junction becomes unstable and may fluctuate back and forth between the vertical characteristic and the expected "two-superconductor" characteristic. With a small increase in current, the junction settles stably on the latter. Expansion of the voltage scale showed some fluctuations even at lower currents in many cases.

One possible explanation that will be suggested is, of course, that in such thin junctions we have not avoided small superconducting shorts across the barrier. There are, however, four experimental points suggesting that this is indeed the Josephson effect.

(1) As pointed out in Josephson's Letter,[1] the effect should be quite sensitive to magnetic fields. Since the proposed proportionality to $\mathrm{Im}(\Delta_{Sn}^{*}\Delta_{Pb})$ is not gauge invariant, we expect an additional dependence on $\sin\int_1^2(2e/\hbar c)\vec{A}\cdot d\vec{l}$, where $\vec{A}$ is the vector potential, which will lead to cancellation of currents in various parts of the barrier if the magnetic flux flowing between the superconductors reaches one or two quanta. With an area of $\sim 10^{-7}$ cm^2, this corresponds to about 1 gauss. We have found (see Fig. 2) that when the junction was carefully shielded by a mu-metal can with a measured interior field of 6×10^{-3} gauss, the vertical characteristic reached 0.65 mA, with no shielding (0.4 gauss), 0.30 mA, and when less than 20 gauss was applied the anomalous behavior was not observed. Fine superconducting filaments should show anomalously high, not low, critical fields.

(2) The effect can only occur if both metals are superconducting and should be proportional to $\sim|\Delta_{Sn}\Delta_{Pb}|$. On cooling the junction we find that the vertical characteristic appears at the tin transition (as measured by the negative resistance region of width Δ_{Sn} first appearing at Δ_{Pb}) within experimental error. It seems unlikely that the "superleaks" would have precisely the same T_c as the tin film.

(3) Critical currents of finely divided specimens never exceed 10^7 A/cm^2. Our observed maximum current of 0.65 mA corresponds, then, to an area A of superleaks of not less than 6×10^{-11} cm^2. Assuming a rather poor conductivity of 10^4 ohm^{-1} cm^{-1} for the filaments, this leads to a conductance, which should be observed even when the filaments are normal, of

$$Y \geq \frac{10^4\times6\times10^{-11}}{10^{-7}} = 6 \text{ mho},$$

whereas (Fig. 2) we observe with an applied field at most 2×10^{-2} mho, most of which is probably thermally excited quasi-particle tunneling.

(4) We attempt to "burn out" the leaks by passing increasing currents through the junction, checking the anomalous current between each step. We observed no change in the vertical characteristic for a number of junctions ($\frac{1}{10}$ to 1 Ω), until (at a voltage usually between 0.3 and 0.6 volt) destruction of the junction resulted. Metallic leaks should burn out before the junction as a whole does.

These last arguments seem to us nearly to exclude the conducting leak hypothesis.

The maximum current we observe is still, even in the best units, about an order of magnitude less than predicted. Other questions concern the fluctuation effects and the fact that thin junctions are necessary to see the effect.

We believe that these questions are probably all best elucidated by looking at the effect as related to a coupling energy between the phases of the gap functions on the two sides. Calculations which will be reported elsewhere show that this energy is proportional to the negative cosine of the phase difference, and in magnitude is

$$\Delta E = (\hbar/e)J_1,$$

where J_1 is the maximum amplitude of Josephson's predicted current. This energy coupling is reduced both by the presence of magnetic fields and by driving current through the unit. In order to observe the dc Josephson effect, this energy must be large enough to keep the phases on the two sides coupled against thermal or other fluctuations.

The total magnitude of ΔE for the entire unit is rather small-of order 1 or a few eV in the thin barriers, 10^{-2} eV in the more normal units. Obviously the thicker units are completely unstable against thermal fluctuations (remember that most of the circuit is at room temperature or higher). The thin barriers are more stable, but as we drive larger currents through them, or apply stray magnetic fields, we lower the coupling energy, and random fluctuations will eventually destroy the dc current and replace it with noise,

A perfectly rigorous way to think of this process is that the energy ΔE serves as a barrier against the passage of quantized flux lines through the unit. It will act as a superconductor only if fluctuations over the applied magnetic stresses cannot drive lines through the barrier region

at an appreciable rate.

We wish to acknowledge the assistance of L. Kopf in the preparation of the tunnel units, and discussions with V. Ambegaokar. B. D. Josephson Informs us that he has independently reached some of the theoretical conclusions of the last few paragraphs.

[1]B. D. Josephson, Phys. Letters 1, 251 (1962).

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MAGNETIC FIELD DEPENDENCE OF THE JOSEPHSON TUNNEL CURRENT

J. M. Rowell
Bell Telephone Laboratories, Murray Hill, New Jersey
(Received 24 July 1963)

When two superconductors are separated by a thin insulating layer (<20Å thick), the form of the tunneling characteristic (current I vs voltage V) can be simply explained in terms of single-particle density-of-states vs energy diagrams.[1] On the basis of these "semiconductor" diagrams the tunneling is regarded as being due to single particles (electrons or holes), and the initial and final density of states enter directly into the tunneling probability for each particle. However, after considering terms others had neglected in the tunneling expression, Josephson[2] predicted that in addition to the "single-particle" currents mentioned above, a second type of tunnel current should be observed. This current is a flow of "Cooper pairs" between the Fermi surface of the two superconductors through the insulating layer. It should be observed as a direct current of limited magnitude flowing through the junction when there is no voltage between the superconductors. When a voltage V is established across the junction, the current alternates at a frequency $2eV/h$. The direct current has been reported previously[3] and arguments presented to differentiate it from currents which might be expected in junctions containing fine metallic shorts. The alternating current, or its interaction with applied microwave fields, has been reported by Shapiro.[4] Reported here is a study of the effect of magnetic fields on the direct current in a number of tunnel junctions of different dimensions. It is found that the current is reduced to minimal values whenever the junction contains integral numbers of flux units and that "self-limiting" affects "cross" junctions with any dimension greater than the "Josephson penetration distance."

The currents were first observed in Sn-I-Pb junctions (Sn evaporated first) but recently more strikingly in Pb-I-Pb junctions. The fabrication of the junctions follows established techniques[5] and the insulator is a thermally grown oxide. The dimensions of the junctions have been varied considerably but the resistance is consistently $\sim 3 \times 10^{-2}$ ohm mm^2 (see Fig. 2). The current voltage characteristics were taken on an X-Y recorder, using a dc supply of much higher resistance than any of the junctions.

The junctions were immersed in liquid helium at 4.2 or 1.3°K, the magnetic field being provided by a copper-wound solenoid surrounded by a Mu-metal can, both immersed in the helium bath. The production of 40 gauss involves negligible heat input. The solenoid was calibrated at room temperature using a Hall effect probe. The Mu-metal shield reduces the field at the sample from other Sources to $<10^{-2}$ G.

Josephson[6] and Ambegaokar and Baratoff[7] have pointed out that the maximum magnitude of the current J, is equal to the flowing when the junction has both films normal and is at a bias $\frac{1}{2}\pi\Delta$. This current will only be observed, however, for a given junction under ideal conditions. As pointed out by Andersion[3,8] and by Ferrell and Prange,[9] the phases of the gap functions on the two sides of the junction are cou-

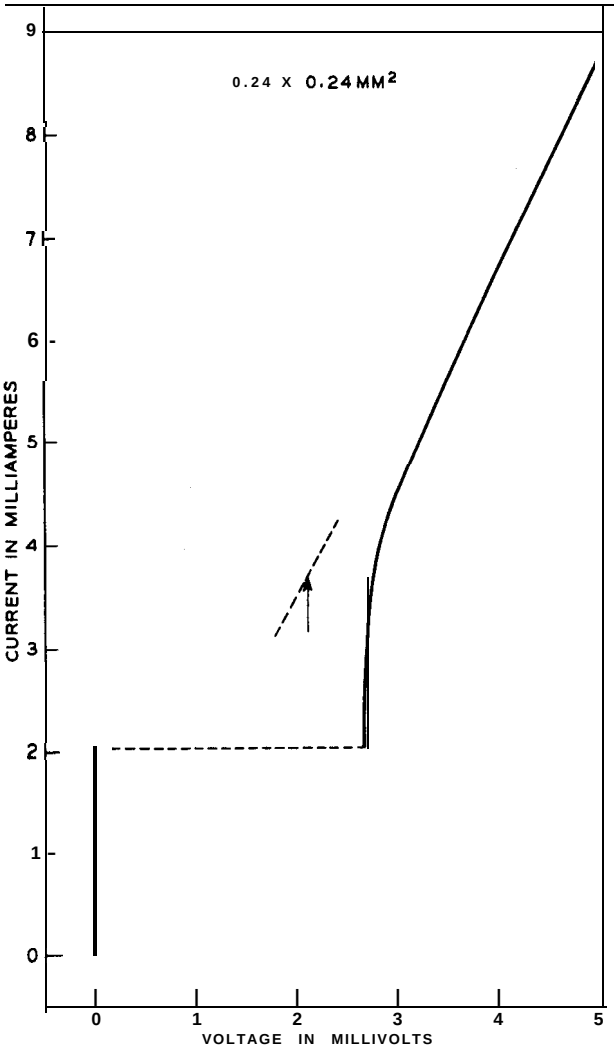
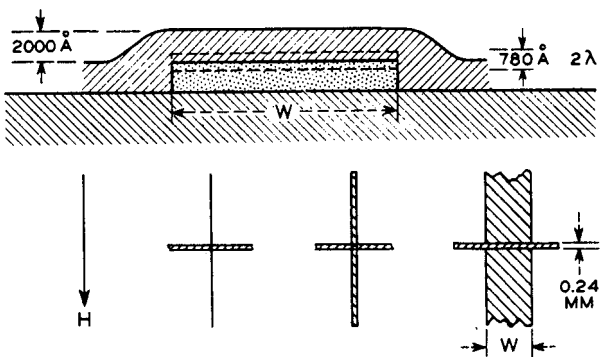


FIG. 1. Current-voltage characteristic for a Pb-I-Pb junction at 1.3°K. The arrow marks the predicted maximum magnitude of the Josephson current.

pled by an energy $hJ_1/2e$. This coupling is reduced by magnetic fields and by current flow through the unit. In order to maintain the flow of Josephson current against thermal and circuit fluctuations, the value of J_1 should be as large as possible. This can be done by making the junction resistance low (the insulator as thin as possible and the junction area large) and by using superconductors with large Δ. The junction area cannot be increased indefinitely, however; apart from the difficulty of making large-area thin insulating layers, the smaller dimension of the junction should remain $\lesssim 2\lambda_J$ where λ_J is the "Josephson penetration distance." This is a consequence[6,9] of the self-magnetic fields generated by the Josephson current, which contain the current flow to within a distance $2\lambda_J$



WIDTH W	0.04 MM	0.24 MM	2.40 MM
FIELD PERIOD	6.5 GAUSS	1.1 GAUSS	0.4 GAUSS
FLUX	2.0×10^{-7} G·CM²	2.1×10^{-7} G·CM²	7.5×10^{-7} G·CM²
I AT $\frac{\pi}{2}\Delta$	0.5 MA	3.7 MA	38 MA
I_0	0.17 MA	2.0 MA	4.5 MA

FIG. 2. Comparison of the dimensions of three junctions and the magnitude and field dependence of the Josephson current.

from the edges of the junction with

$$\lambda_J = (16\pi\lambda e^2 J_1/hc^2)^{-1/2},$$

where λ is the penetration depth, 390 Å for lead films.[10]

The unit in which the ratio of the observed Josephson current I_0 to the maximum magnitude (I at $\frac{1}{2}\pi\Delta$) was largest was one of dimensions 0.24 X 0.24 mm; as can be seen from the current-voltage characteristics of Fig. 1, this ratio is 0.55. Figure 2 gives the relevant data for two other junctions. The one of smaller area had the ratio 0.35, whereas the other, of much larger area, had 0.1. This is surprising at first sight as one film of this unit is still only 0.24 mm wide ($\sim\lambda_J$), and self-limiting of the current should not be serious. The junction is not, however, identical to the idealized model used by Ferrell and Prange[9] in which current flow in the two superconducting films is parallel and opposite. In the "cross" junction of Fig. 2 the current flows along the top narrow film down through the junction and away from the junction along the lower wide film in a direction perpendicular to the current flow in the top film. As the current density will be greatest in the narrow film, the field in the junction will be generated almost

solely in the direction and plane of the wide film. If the field due to current in the wide film is assumed zero, then the new penetration depth is $\sqrt{2}\lambda_J$ and current is confined to a width $2\sqrt{2}\lambda_J$ of the junction. For this junction λ_J is calculated as 0.22 mm, $2\sqrt{2}\lambda_J = 0.62$ mm. This indicates that only one quarter of the junction is carrying current and explains the low observed current of 4.5 mA.

It has been observed by Josephson[6] that the effect of an external magnetic field (or a current flowing along one film of the junction) would be to reduce the direct current to a minimum whenever the junction contained integral numbers of flux units ($hc/2e = 2.1 \times 10^{-7}$ gauss cm^2). We have observed this effect in junctions of various dimensions, but it is most striking when the film along the field direction is as narrow as possible. Figure 3 shows the variation of observed current I_0 as a function of magnetic field for a Pb-I-Pb junction made of films 0.040 and 0.24 mm wide, the 0.040-mm wide film being roughly along the field direction. When a field of 6.5 gauss is applied, the Josephson current is reduced by a factor >600 and cannot be measured with the existing experimental sensitivity. At 13.0 and 19.5 gauss, the current again goes through minima with successively decreasing maxima between. The area of junction containing flux is the width $W \times 2\lambda = 3.1 \times 10^{-8}$ cm^2 (see cross section of junction, Fig. 2). Thus a field of 6.5 gauss corresponds to a flux of 2.0×10^{-7} gauss cm^2 in the junction, which is indeed the flux unit. Considering screening and demagnetizing effects of the films, it is surprising that such a good value is obtained. A larger junction, 0.24 x 0.24 mm, had a period of 1.1 gauss, but the minima were not so well defined as the current did not reduce so effectively to zero. Again this corresponds to one flux unit being in the junction. For the large-area junction, 2.4 x 0.24 mm, the minima were poorly defined with approximately a 0.4-gauss period. If the full width of the junction is effective in carrying current, one obtains the flux as 7.5×10^{-7} gauss cm^2, but if $2\sqrt{2}\lambda_J$ is effective then the flux is 1.9×10^{-7} gauss cm^2. This appears to confirm the conclusion above that the Josephson current flow is confined to a width $2\sqrt{2}\lambda_J$ at the edge of the junction.

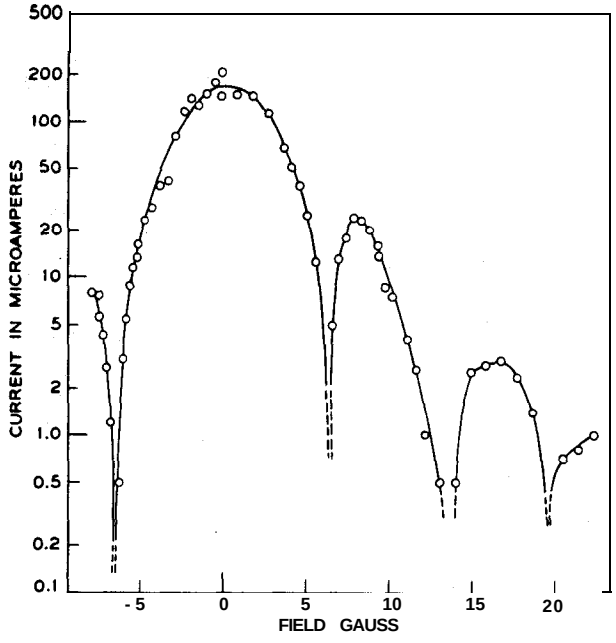


FIG. 3. The field dependence of the Josephson current in a Pb-I-Pb junction at 1.3°K.

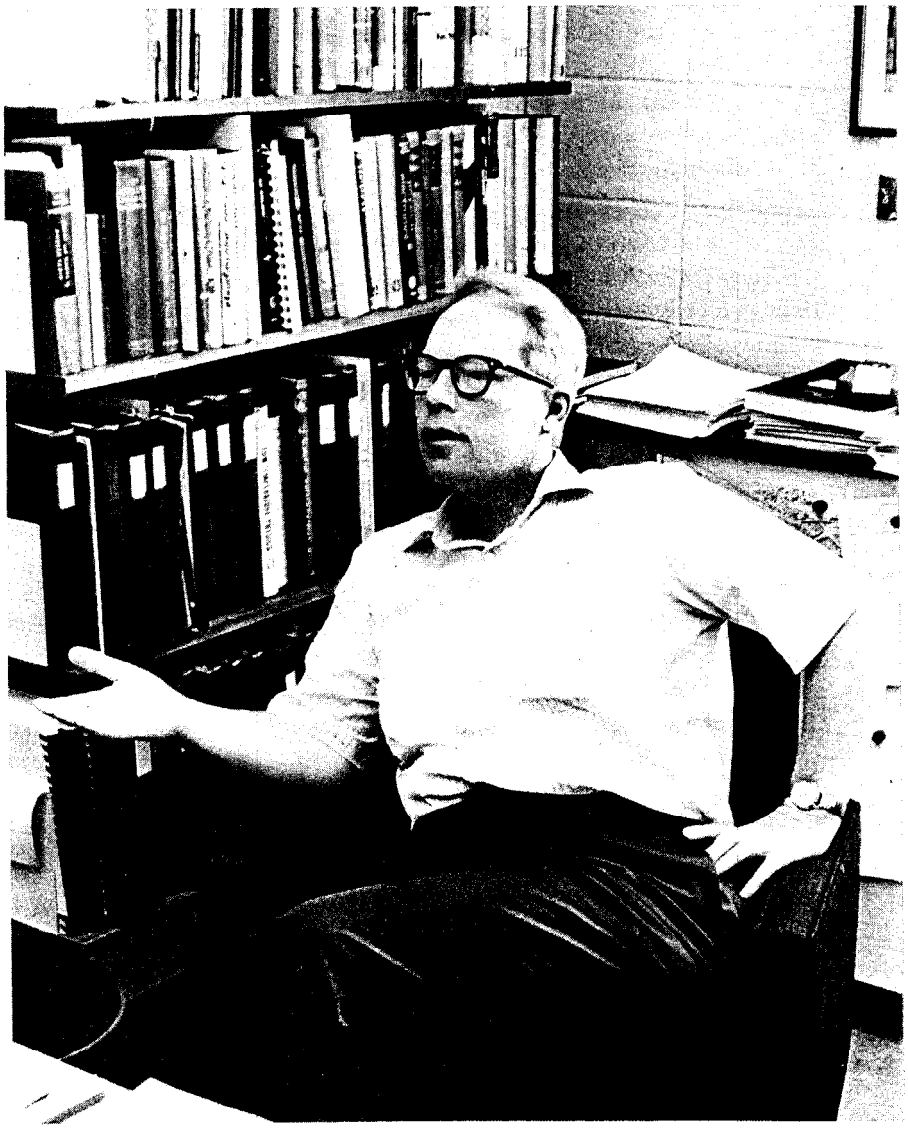
I wish to acknowledge many valuable discussions with P. W. Anderson and the assistance of L. Kopf in the preparation of the tunnel units.

[1] I. Giaever, Phys. Rev. Letters 5, 464 (1960); J. Nicol, S. Shapiro, and P. H. Smith, Phys. Rev. Letters 5, 461 (1960).
[2] B. D. Josephson, Phys. Letters 1, 251 (1962).
[3] P. W. Anderson and J. M. Rowell, Phys. Rev. Letters 10, 230 (1963).
[4] S. Shapiro, Phys. Rev. Letters 11, 80 (1963).
[5] J. C. Fisher and I. Giaever, J. Appl. Phys. 32, 172 (1961).
[6] B. D. Josephson (private communication).
[7] V. Ambegaokar and A. Baratoff, Phys. Rev. Letters 10, 486 (1963). In a correction to this. Letter [Phys. Rev. Letters 11, 104(E) (1963)], the authors point out that the temperature dependence of J_1 for a symmetrical junction is given by

$$J_1 = \tfrac{1}{2}\pi R_N^{-1}\Delta(T)\tanh\tfrac{1}{2}\beta\Delta(T),$$

but for Pb-I-Pb at 1.3°K this correction is negligible.
[8] P. W. Anderson, Proceedings of 1963 International Spring School of Physics, Ravello (to be published).
[9] R. A. Ferrell and R. E. Prange, Phys. Rev. Letters 10, 479 (1963).
[10] J. M. Lock, Proc. Roy. Soc. (London) 208, 391 (1951).

DISCOVERY STORY I



Sidney Shapiro in his office at the University of Rochester [Photo: Alan G. Knapp]

One Researcher's Personal Account

by Sidney Shapiro

IN 1963, Anderson and Rowell* experimentally verified a theoretical prediction made by Brian Josephson** the previous year-that one could have a net current flowing across a Josephson junction (two superconductors separated by a thin layer of insulating material) if there is a zero voltage across the junction. This current vanishes as soon as a voltage is placed across the junction. The results of Anderson and Rowell made it quite obvious to all of us

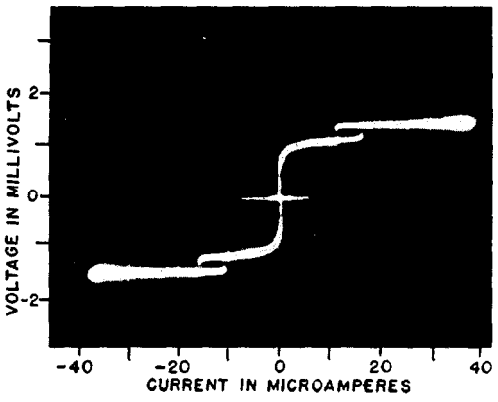


Fig. 1
Voltage vs tunneling current for an Al-Al_2 0_3 -Pb sandwich at 1° K. From Nicol, J., Shapiro, S., and Smith, P., *Phys. Rev. Letters* 5, 461 (1960).

working with superconducting tunnel junctions that those annoying 'shorts' we frequently observed were actually, in many cases, the zero voltage Josephson current. [See, for example, *Phys. Rev. Letters* 5, 461 (1960), Figure 1.] In my own laboratory at Arthur D. Little, Inc., we were soon waving magnets around, 'proving' to ourselves that our suspicions were correct.

I had been working for some time on a program sponsored by the Air Force to explore the potential of the sharply nonlinear voltage-current (V-I) curve associated with superconducting tunneling. Experimental observations had been completed at low radio frequencies, so 1 had transferred my attention to examining the responses of various junctions to microwaves in the X-band (about 9GHz). I was especially interested in what modifications might appear in the V-I curves under intense microwave radiation.

My two assistants in this project were junior professional personnel: Andre Janus and Sandor Holly. Andre was primarily occupied with operating the vacuum evaporation equipment used in the delicate task of preparing the junctions: he also assisted Sandor and me in making experimental observations.

One afternoon, as he was leaving for the day, Andre reported that there was still a considerable amount of liquid helium left in the dewar from the day's run, in case I was interested in taking more data. (The Josephson junction must be cooled to a few degrees above absolute zero in order to make the junction superconducting.) It had become almost routine for Andre to drop by with this information and I usually responded by taking more data. What made this particular afternoon different from the others was Andre's additional comment that this sample seemed to have a rather large Josephson current. We had been irradiating the junction with microwaves of a known frequency, in order to induce a series of steps in the V-I curve, so I asked Andre if he had seen whether the microwaves had any effect on the Josephson current. He replied that he hadn't, since he had been asked to take data of a different sort.

A Strange Effect

When Andre had gone, I went into the laboratory intending to check only a few points, including the Josephson behavior of the sample. Within a few minutes, I had observed the rf-induced steps in the Josephson current. At the time, I wasn't sure exactly what I was observing; but I knew that something strange, wonderful and exciting was going on! (It was some time later that I realized that the steps I had observed satisfied the relation predicted: $nf = 2eV/h$.)

I took what data I could by photographing V-I curves at various settings of the incident microwave power and also varied the temperature somewhat. Unfortunately, it was not possible at that time to vary the frequency of the radiation because the sample was mounted in a resonant cavity.

I remember becoming suddenly concerned that the sample's zero voltage current might not actually be due to the dc Josephson effect, but might be simply a 'short.' There were several available methods for checking this. I chose the method of using large amplitude ac sweeps for the oscilloscope display to show that the observed current was insensitive to the power dissipated in the sample. To my delight, this proved to be the case. The effect was real!

The next few days were hectic. Andre, Sandor and I continued to experiment in order to be certain that the effect I had observed was reproducible, first on the same tunnel junction and then, when it soon died (as they all rapidly did in those days), on others. Simultaneously, we were trying to understand what our observations meant!

It immediately became clear that the effects were reproducible, but, as often is the case when one contends with Nature, it took many tries with many junctions before we were able to obtain results as good as with the first junction. As should be obvious from our Letter†, we were totally fascinated by the phenomena and not at all interested in the accompanying negative resistance and noise effects.

As our understanding grew, we knew we had to repeat the experiment at another widely separated frequency. This we could do without appreciably changing the experimental setup-by taking advantage of the multiple resonance character of our microwave cavity. Although only one mode lay within the X-band, we were able to find a higher-order mode that fell within the range of an available K-band (∿ 25 GHz) klystron. To input the energy, we utilized the fortunate coincidence that a standard X-band wave-guide has a *height* that is nearly the same as the *width* of a standard K-band wave-guide. Hence we could be sure that only the fundamental K-band wave-guide mode would propagate in our experimental X-band guide by transforming the width of the former into the height of the latter. Sandor easily built such a piece of pipe in one day. We were then able to see the rf-induced steps in the K-band as well as in the X-band. It was then simple to confirm the expected proportionality between the frequency and the step separation in voltage.

Cont. p. 61

* **Anderson, P.W., and Rowell, J.M., *Phys. Rev. Letters* 10, 230 (1963).**

** **Josephson, B.D., *Phys. Letters* 1, 251 (1962).**

† **Shapiro, S., *Phys. Rev. Letters* 11, 80 (1963). Reproduced on the following page.**

JOSEPHSON CURRENTS IN SUPERCONDUCTING TUNNELING: THE EFFECT OF MICROWAVES AND OTHER OBSERVATIONS*

Sidney Shapiro
Arthur D. Little, Inc., Cambridge, Massachusetts
(Received 13 June 1963)

In the course of experiments on the effect of microwave fields on superconducting tunneling, we have had occasion over the past few months to fabricate many tunneling crossings of low resistance (5-20 Ω with a crossing area of 1.5 $\times 10^{-4}$ cm^2). Every one of these samples has exhibited the zero-voltage currents predicted by Josephson[1] and attributed, in effect, to the tunneling of Cooper pairs. The observation of. these currents has already been reported by Anderson and Rowell.[2] Our experiments have brought to light several new effects which we summarize below.

The samples were $Al/Al_2O_3/Sn$. Two five-mil-wide Al lines, evaporated onto cleaned glass substrates, were oxidized in a glow discharge generated in. an atmosphere of about 0.1 Torr of dry oxygen for 15 seconds. A five-mil-wide cross-strip of Sn was then evaporated forming two samples on each substrate.

The tunneling current versus voltage characteristics were displayed on an X-Y oscilloscope. A low-impedance source was used to drive the loop containing the sample and the current-measuring resistor. The latter, and thus the circuit load line, could be varied, either in calibrated steps or continuously, from 10 Ω to 10000 Ω. The source was either dc, ac, or both in combination. Generally 60-cps ac was used though other frequencies were employed as desired. No attempt was made to shield the earth's magnetic field. Most data were taken at about 0.9°K.

The following observations were noted in the course of experiments with a large number of tunneling crossings:

1. Using an ac display, Fig. 1 shows for a typical sample the zero-voltage current predicted by Josephson and previously observed with a dc technique by Anderson and Rowell. During each half-cycle of the sweep, current

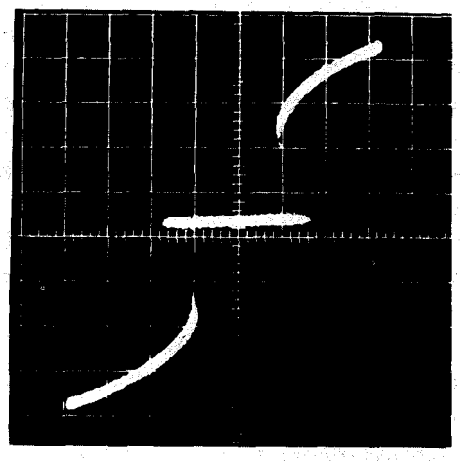


FIG. 1. I-V characteristic near origin showing zero-voltage Josephson current and negative resistance switching trace. Vertical scale 58.8 μV/cm, horizontal scale 130 nA/cm.

flows through the sample without developing any voltage across it until, at a current which depends upon temperature, crossing resistance, and circuit load line, switching to the familiar single-particle tunneling characteristic occurs along the circuit load line. The switching characteristic is stable and reproducible, and is followed both when the current is increasing and when it is decreasing. It is indicative of a negative resistance.

2. Other investigators[3] in this laboratory have observed rf oscillations associated with this region. The samples, placed in appropriately designed oscillator circuits, produced rf at the circuit resonant frequency of about 100 Mc/sec. In view of their frequency, these oscillations occur at too high dc voltages to be the ac supercurrents also predicted by Josephson.[1]

The rf oscillations appear to confirm that the Josephson current region decays toward the single-particle tunneling characteristic by means of a negative resistance, despite the fact that we have not been able to trace out any appreciable portion of the negative resistance.

3. The characteristic as shown in Fig. 1 can be traced out with dc and follows exactly the course of the ac display. On occasion, however, much more ac can be carried at zero voltage than dc; fifteen times as much or more is not uncommon. The same ac trace is obtained for all sweep frequencies employed (8 to 800 cps). Electrical transients can cause the excess ac to disappear and to reappear, but they have no effect on the rest of the characteristic. Frequently, the excess current is noisy in character, i.e., the amplitude of ac carried before switching takes place varies randomly from cycle to cycle. The excess ac appears only at temperatures well below that at which the zero-voltage current is first observed, and is quenched by lower dc magnetic fields and microwave fields than affect noticeably the stable zero-voltage current. We have no explanation for the excess ac and can only conjecture that it may be related to the ac effect predicted by Josephson.

The effect of microwave power in modifying the *I-V* curve, especially the zero-voltage currents, was also studied. The samples were mounted in a microwave cavity resonant, at low temperatures, at about 9300 Mc/sec and 24 850 Mc/sec. All the following observations were independent of sweep frequency and were also seen when dc was passed through the sample.

1. Figure 2 shows for a typical sample the initial effect of 9300-Mc/sec microwave power on the *I-V* characteristic. Without applied power, the trace (top) is similar to that of Fig. 1. With a few tens of microwatts applied, however, the zero-voltage currents become noisy and gradually vanish resulting in regions of zero slope (or almost zero slope) in which the current rises at (or almost at) fixed voltage across the sample. The voltage at which the zero-slope regions occur is equal to $\pm h\nu/2e$, where ν is the microwave frequency. Note that the origin, i.e., the point of zero voltage and zero current, is not a stable point. As more power is applied (not shown), further zero-slope regions appear at still higher voltage, and the origin alternates between being stable and unstable. The current amplitude in a given zero-slope region varies periodically with increasing power. The interval in voltage from one zero-slope region to the next is not always $h\nu/2e$; sometimes a step is missing so that the voltage interval is $h\nu/e$.

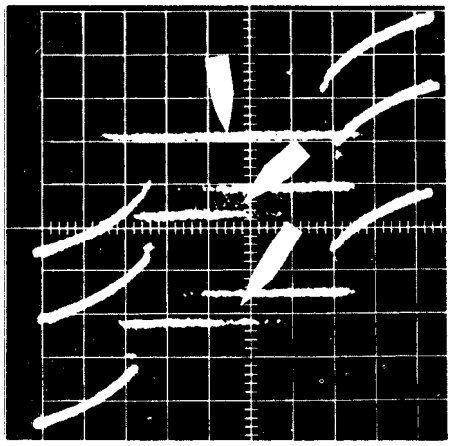


FIG. 2. Initial effect of microwave power. Pointers mark origin, which becomes noisy and vanishes as zero-slope regions at $\pm h\nu/2e$ appear. Note negative resistance at origin. Vertical scale 58.8 μV/cm, horizontal scale 13 nA/cm.

2. Figure 2 demonstrates still another startling feature of the effect of microwaves on the Josephson currents. In the bottom trace the origin has vanished as a possible state, and the system is able to remain biased at, e.g., $+h\nu/2e$, not only when positive current is flowing but even when the current is zero and, more astonishingly, even when the current is reversed and made negative. Finally, at some value of negative current, the voltage switches to $-h\nu/2e$. Thus, the zero-slope regions which occur with voltage $+h\nu/2e$ across the sample are connected by what ap-

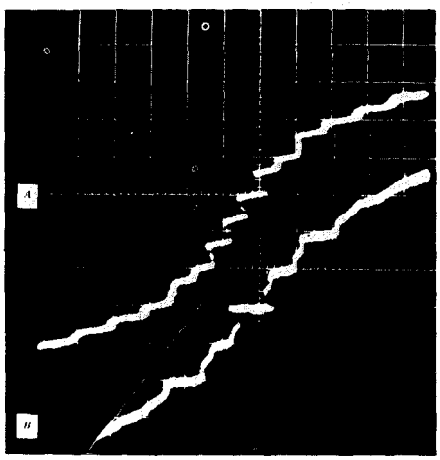


FIG. 3. Microwave power at 9300 Mc/sec (*A*) and 24850 Mc/sec (*B*) produces many zero-slope regions spaced at $h\nu/2e$ or $h\nu/e$. For *A*, $h\nu/e$ = 38.5 μV, and for *B*, 103 μV. For *A*, vertical scale is 58.8 μV/cm, horizontal scale is 67 nA/cm; for *B*, vertical scale is 50 μV/cm, horizontal scale is 50 μA/cm.

pears to be a negative resistance at the origin. Negative-resistance-like regions occur between virtually every pair of zero-slope regions. At power levels of about ten milliwatts, these negative-resistance-like regions gradually vanish (the sample can be biased to any voltage), and the "zero-slope" regions exhibit a finite slope. Both these high-power effects are probably due to the superimposed conventional detection processes associated with the background single-particle tunneling curve.

3. Similar effects occur at 24 000 Mc/sec. Figure 3 shows the numerous steps at 9300 Mc/sec (A) and at 24 850 Mc/sec (B) which are present at intermediate microwave power levels.

Josephson[1] has already discussed briefly the effect rf should have on the pair-tunneling supercurrent. He predicted the occurrence of zero-slope regions separated by $h\nu/2e$ in the *I-V* characteristic in the presence of the rf field. This prediction was based on the frequency modulation by the rf of the ac supercurrents previously referred to. Our experiments have confirmed this prediction and represent indirect proof of the reality of Josephson's ac supercurrent. In addition, they have brought to light several startling accompaniments of the Josephson effect, namely, the instability of the origin; the dc negative resistance at the origin; the noise associated with the onset of instability; and the periodic change at a given voltage level, e.g., at zero voltage, between stable and unstable as the microwave power changes.

We gratefully acknowledge the assistance of Mr. Andre R. Janus and helpful discussions with Mr. Sandor Holly.

*Supported by the Electronic Technology Laboratory, Aeronautical Systems Division, U. S. Air Force, Wright-Patterson Air Force Base, Ohio, under Contract No. AF33(657)-7915.

[1]B. D. Josephson, Phys. Letters 1, 251 (1962).

[2]P. W. Anderson and J. M. Rowell, Phys. Rev. Letters 10, 230 (1963).

[3]W. Schonbein and P. H. Smith (private communication).

cont. from p. 57

It was clear to us right away that we had observed something connected with the Josephson effect, but we weren't sure just what that 'something' was! Josephson's original paper had been until then (and remains today!) largely unintelligible to me. It soon became apparent, however, that the steps we observed were connected with the ac Josephson effect and that we had unwittingly carried out exactly the experiment that Josephson describes in one small paragraph of his paper! From that brief description we were able to extract the prediction of constant voltage steps: they should appear at the voltages $V = n(hf/2e)$. However, it wasn't until Josephson wrote me a short letter, long after the publication of my paper, that I finally understood his point about the frequency modulation of the ac supercurrents and the significance of the microwave power dependence of the step amplitudes. §

‡ For a version of this experiment suitable for an undergraduate laboratory see: Richards, P. L., Shapiro, S., and Griner, C., *Amer. Jour. Phys.* **36,** 690 (1968).

§ Additional results were presented at the Colgate Superconductivity Conference in the summer of 1963 and were later published: Shapiro, S., Janus, A., and Holly, S., *Rev. Mod. Phys.* **36,** 223 (1964).

Results Quickly Accepted and Extended

The experimental results were quickly accepted by the physics community. The actual experiment can be done so simply‡ at any laboratory able to make and irradiate Josephson junctions that no special confirmation of the results seemed necessary; nor, to the best of my knowledge, was any ever published.

Many colleagues privately reported repeating the experiment– and they rapidly extended it in new directions. The most noteworthy of these developments were:

(1) the extension of the effect to weak links of various forms, primarily by Dayem[1] of Bell Laboratories and Lambe[2] of the Ford Scientific Laboratory

(2) the observation and use of self-resonant modes in tunnel junctions to directly observe radiation produced by the ac Josephson currents. This was done first by a group in the Soviet Union and later by a group at the University of Pennsylvania[3]

(3) theoretical work by Riede1[4] and Werthamer[5] who predicted a frequency (voltage) dependance of the Josephson current amplitude (the 'Riedel peak'). This prediction was experimentally confirmed by Hamilton[6] and Buckner, Finnegan and Langenberg[7]

(4) the use of the rf-induced step phenomenon by the Pennsylvania group to measure precisely the value e/h (see Langenberg's Discovery Story in this chapter) and its subsequent use by the National Bureau of Standards as a new standard of voltage.

1. Anderson, P.W., and Dayem, A.H., *Phys. Rev. Letters* **13,** 195 (1964).
2. Lambe, J. *et. al. Phys. Letters* **11,** 16 (1964).
3. Langenberg, D., Scalapino, D., and Taylor, B., *Proc. I.E.E.E.* **54,** 560 (1966).
4. Riedel, E., Z. *Naturforsch.* **19a,** 1634 (1964).
5. Werthamer, N.R., *Phys. Rev.* **147,** 225 (1966).
6. Hamilton, C.A., *Phys. Rev. B* **5,** 912 (1972).
7. Buckner, S., Finnegan, T., Langenburg, D., *Phys. Rev. Letters* **28,** 150 (1972).

Copenhagen, March 1973

A portion of Discovery Story I by Sidney Shapiro was taken from a letter written for George L. Trigg's study "Landmark Experiments in 20th Century Physics."

EXPLANATORY PHYSICS NOTES FOR NON-SPECIALISTS

The Josephson Junction

A pair of superconductors, separated by a thin barrier, forms what is generally called a 'Josephson junction.'

A Josephson junction can be viewed rather generally as a 'weak' spot in a superconducting system at which a localized difference in quantum phase can be established between two superconducting regions. This phase difference can be monitored by observing the current-referred to as the 'supercurrent' -through the junction. This supercurrent responds to electric and magnetic fields and it is this response which leads to a variety of interesting and useful Josephson phenomena.

Josephson junctions can be constructed in a number of ways. The most important varieties in which the necessary weak coupling has been achieved are shown in Fig. 1 and described in its captions on the page opposite.

Behavior of the Josephson Supercurrent

The behavior of the supercurrent across a Josephson junction is described by two basic equations:

$$j = j_1 \sin\left(\phi - \frac{2e}{\hbar c}\int A \cdot dr\right) \qquad (1)$$

$$\partial\phi/\partial t = 2\Delta\mu/\hbar \qquad (2)$$

where ϕ is the difference in quantum phase across the junction (i.e., between the two superconductors), j is the junction supercurrent density, A is the magnetic vector potential, and $\Delta\mu$ is the electrochemical potential difference across the junction. The maximum value of j, j_1, depends on the material used in the junction, the temperature and the voltage across it. For a given junction at a fixed temperature, j_1 may be considered constant. The major contribution to the electrochemical potential difference is the electrostatic potential eV. The second equation can thus be rewritten

$$\partial\varphi/\partial t = 2eV/\hbar \qquad (3)$$

If V = 0 and A = 0, the two equations predict a zero-voltage supercurrent of maximum density = j_1. If V $\neq$ 0 but constant, an oscillating (ac) supercurrent is predicted. This ac supercurrent will have a characteristic frequency:

$$\nu = 2\Delta\mu/h = 2eV/h \qquad (4)$$

Cont. p. 64

LOAN TIME
-7 DAYS-

Figure 1.

(a) *A tunnel junction of the type originally considered by Josephson. Two superconducting films are coupled via coherent tunneling of bound Cooper pairs (a pair of electrons which behaves as one 'particle') through an insulating oxide film between the layers.*

(b) *A Dayem bridge or 'weak link'junction in which the two superconductors are coupled by a narrow constriction. In order to be coupled weakly enough to exhibit Josephson behavior, the bridge must be extremely narrow. Due to additional constraints-one of which is the technical ability to construct such a narrow bridge-this junction can only be operated in a narrow range of temperature just below the superconducting transition temperature, T_c.*

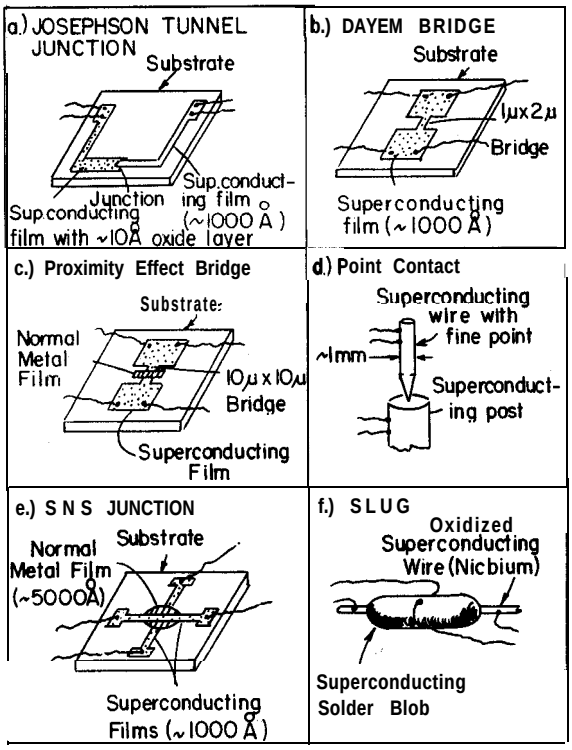
(c) *Proximity effect bridge. By overlaying a metal film in a normal state of conductivity, the superconductivity of the link is weakened. This produces the same effect as maintaining the Dayem bridge near T_c. The link can thus be made larger and is easier to construct from a practical viewpoint.*

(d) *Point contact junction in which the junction characteristics are varied by adjusting the pressure of the contact. This junction is extremely easy to construct. Tunneling can also be observed by plating the lower post with a thin oxide layer, as in a simple tunnel junction seen in Fig. 1 (a).*

(e) *Superconductor-normal metal-superconductor. Two superconducting films are separated by a thick (10^3 to 10^4 Å) metal layer. A weak superconductivity is induced in the metal by the superconductors and leads to the necessary weak coupling. This type was used by Langenberg and collaborators, (Fig. 4).*

(f) *Slug junction. An oxidized niobium wire is surrounded by a blob of ordinary solder. Multiple junctions form between the niobium and solder via small bridges through the oxide. This device exhibits complex behavior combining weak linkage and tunneling effects.*

Figure 1. *The most common types of Josephson junctions.*



Cont. from p. 62

This last expression is the Josephson frequency-voltage relation and is the key to the use of the Josephson effect in determining the fundamental constant e/h.

While the sinusoidal form of the first equation accounts rather well for nearly all observed Josephson phenomena, it is too simple to provide an exact description of the current-phase relationship, particularly when certain types of weak links are considered. Fortunately, this equation is not essential in the use of the Josephson effect in determining values of the fundamental constants.

The nonlinearity of Equation 1 does lead, however, to a rich variety of Josephson phenomena, which cannot be considered here. We show in Fig. 2 dc current-voltage characteristics for several typical junction types. The sharp increase at 2.7 mV in 2(a) and 2(b) is not due to a Josephson phenomenon, but is due to single particle tunneling.

The zero-voltage current is quite apparent in all three curves. The current component due to the flowing of normal quasi-particles at small nonzero voltages is negligible in tunnel junctions, but can be seen to be important in weak links. A dc supercurrent may also appear at nonzero voltages. It decreases monotonically with voltage and is merely the dc component of the periodic, but nonsinusoidal, ac supercurrent.

An additional effect occurs in tunnel junctions. The junction acts as a short section of parallel plate transmission line and can support resonant electromagnetic modes. In large junctions of this type, the ac supercurrent couples to these resonant modes producing the singularities shown in Fig. 2.

The ac supercurrent manifests itself experimentally in two distinct ways which can be used to determine e/h. In a voltage-biased junction, the supercurrent emits coherent electromagnetic waves. While this phenomenon has been used (by measuring the frequency of emitted radiation) to measure e/h, the low power levels (10^{-11} watts) and limits on usable bias voltages make a second effect more convenient. If a Josephson junction is exposed to an rf field, well-defined current steps appear in the dc I-V current characteristic. These steps appear at discrete voltages $V_n = nh\nu/2e$ where ν is the rf frequency and n is an integer (see Fig. 3). The steps result from the beating of the supercurrent with the harmonics of the rf current induced in the junction by the incident radiation. This latter effect was the primary one used by Langenberg and coworkers in their sub-part-per-million-accuracy determination of e/h.

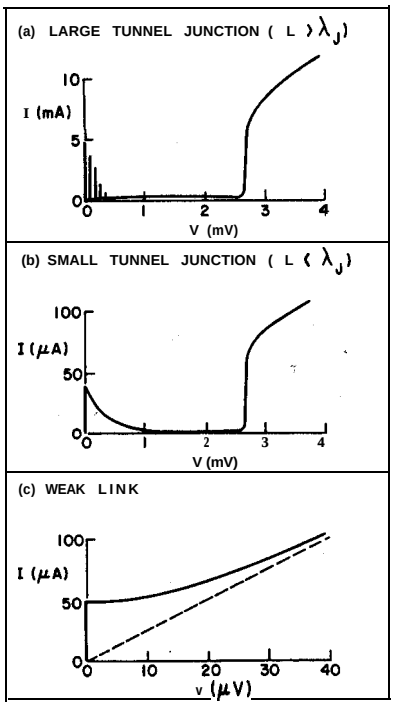


Fig. 2. *Typical Josephson junction current-voltage characteristics.*
λ_J here is the Josephson penetration depth

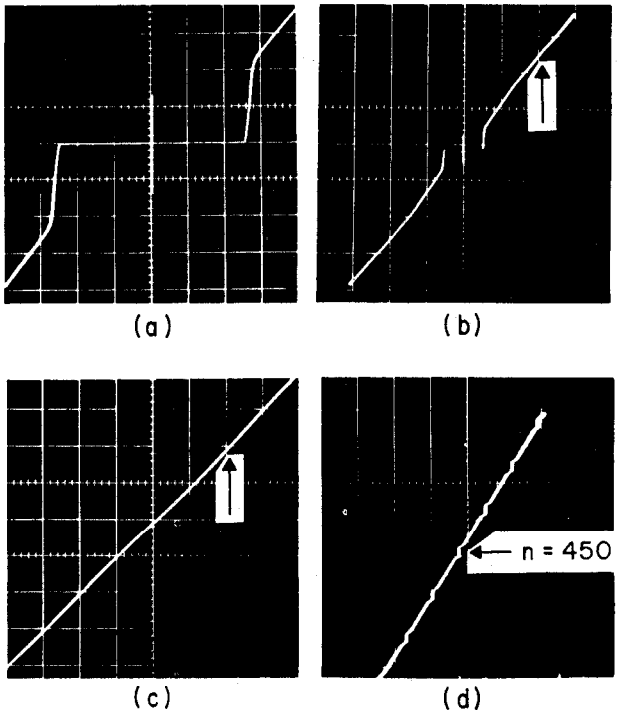
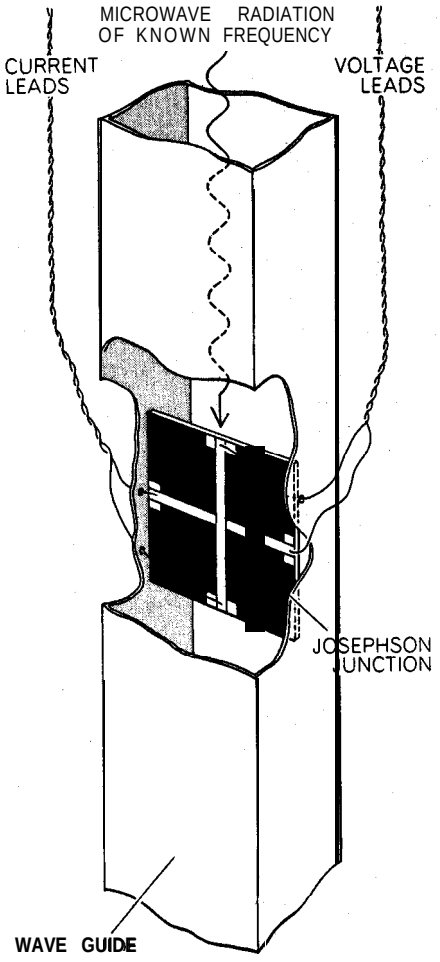


Fig. 3. (a) *The current-voltage characteristic of a* **Pb–PbO_x–**Pb *Josephson tunnel junction.*
Current is vertical with a scale of 2.5 mA/cm; voltage is horizontal with a scale of 1 mV/cm.
(b) *The same characteristic with current scale 10 mA/cm and voltage scale 5 mV/cm.*
The arrow just above 10 mV marks the site of (d).
(c) *The characteristic with 11 GHz microwave radiation incident.*
The scales and the arrow are the same as in (b).
(d) *Microwave-induced Shapiro steps above 10 mV.*
The current scale is 50 μA/cm and the voltage scale is 25 μV/cm.



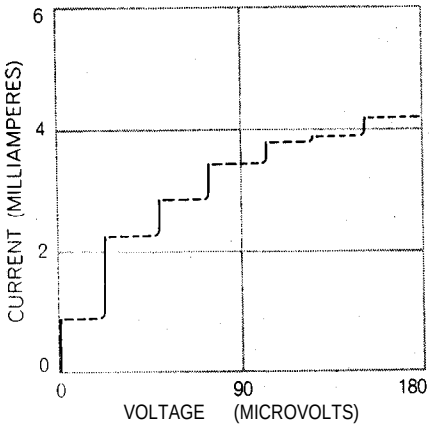


Fig. 4. *Microwave iwadiation of a Josephson junction mounted inside a wave guide induces a series of steps in the current vs voltage graph which is characteristic of the junction being examined The junction can then be biased (current leads) onto one of these induced steps and the step voltage measured with a potentiometer (voltage leads). The value of e/h can then be determined since the microwave frequency is proportional to the step voltage, with a proportionality constant = 2e/h.*

EXPLANATORY PHYSICS NOTES FOR NON-SPECIALISTS

Determination of α from Measurement of e/h

The gyromagnetic ratio of the proton is defined as the ratio of the angular precession frequency of a proton in a magnetic field B to the magnitude of the field. Thus,

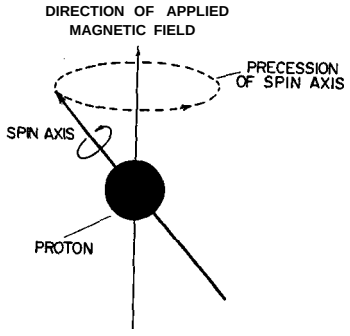
$$\gamma_p = \omega_p / B.$$

(Note that since $\hbar\omega_p = 2\mu_p B$, $\gamma_p = 2\mu_p / \hbar$.) γ_p is a most important input datum because, being the largest source of error, in combination with the Josephson-effect value of e/h, it essentially determines our WQED* value of the fine structure constant. Two different techniques have been used for measuring γ_p. The more accurate method involves the free precession of protons, usually in a spherical water sample, in a small magnetic field of the order of 0.001 T. This field is obtained by passing a current known in terms of as-maintained electrical units through a precision solenoid; the field is then calculated from the current and the accurately known dimensions of the solenoid. Most of the Earth's magnetic field is usually bucked out by-large external Helmholtz coils, and any residual component of the Earth's field along the axis of the solenoid is averaged out by reversing the current in the solenoid and remeasuring ω_p. The residual component normal to the solenoid field is usually so small that it can be neglected. (It typically contributes an uncertainty of about 0.1 ppm.) In order to raise the signal-to-noise level to the point where an accurate measurement of ω_p can be made, the water sample is usually prepolarized in a large magnetic field. The magnetization produced in the sample by this polarization is oriented normal to the solenoid field by the application of an rf pulse and then freely precesses around it, inducing a signal in a pickup coil at the frequency ω_p.

The second method uses resonant absorption and is usually carried out in a conventional electromagnet at a much higher field (of order 0.5 T). In this technique, the proton sample (usually cylindrical in shape) is located inside a small coil with axis normal to the field. An rf signal at a frequency ω_p is applied to the coil and resonant absorption of energy by the protons is detected by means of a change in the balance of an rf bridge containing the coil. In the high-field method, the magnetic field is much more difficult to measure than in the low-field method, and its determination is the major source of error.

For a more detailed discussion of the two methods see: B.N. Taylor, W.H. Parker and D.N. Langenberg, "Determination of e/h, QED and Fundamental Constants," *Rev. Mod Phys.* 41, 375 (1969).

**WQED = without quantum electrodynamics.*



Spin precession of the proton about the direction of applied magnetic field. The ratio of the precession frequency to the strength of the applied field is the gyromagnetic ratio of the proton (γ_p).

NAME OF CONSTANT	SYMBOL	UNITS	VALUE	± UNCERTAINTY PARTS/MILLION
VELOCITY OF LIGHT IN VACUUM	c	10^8 METERS PER SECOND	2.9979250	±0.33
CHARGE OF THE ELECTRON	e	10^{-19} COULOMB	1.6021917	±4.4
PLANK'S CONSTANT	h	10^{-34} JOULE-SECOND	6.626196	±7.6
MASS OF ELECTRON	m_e	10^{-31} KILOGRAM	9.109558	±6.0
INVERSE OF FINE-STRUCTURE CONSTANT	α^{-1}	1 (PURE BUMBER)	137.03602	± 1.5
RYDBERG CONSTANT	R_∞	10^7 PER METER	1.09737312	±0.10

The new value for the fine-structure constant, α, obtained in the experiment described in the forthcoming pages, produced changes in several other constants which depended on it. Recent experiments have also made minor changes in several other fundamental constants. Above are several important physical constants and their presently accepted values, along with the associated uncertainty in parts per million.

Relation Between α and e/h

A value of the fine-structure constant can be derived from the value of 2e/h and other directly measured quantities by use of the equation

$$\alpha^{-1} = \left(\frac{c \ \mu_p \ 2e}{4 R_\infty \ \gamma_p \ \mu_B \ h} \right)^{1/2}$$

where c is the velocity of light, R_∞ is the Rydberg constant for infinite mass, γ_p is the gyromagnetic ratio of the proton, and μ_p / μ_B is the magnetic moment of the proton in units of the Bohr magneton. The best values for these quantities are: c = 2.997925 x 10^8 m sec^{-1} ± 0.3 ppm, R_∞ = 1.0973731 x 10^7 m^{-1} ± 0.1 ppm, μ_p / μ_B = 1.52103264 x 10^{-3} ± 0.3 ppm, γ_p = 2.6751965 x 10^8 rad sec^{-1} T^{-1} ± 3 ppm, and 2e/h = 4.83593718 x 10^{14} Hz V^{-1} ± 0.12 ppm.

Fine-Structure Constant

$$\alpha = \frac{e^2}{\hbar c}$$

One Researcher's Personal Account

by
Donald Langenberg

QED* Discrepancy

At the time this experiment was performed, there existed a discrepancy between the value of the fine structure constant, α, obtained by measuring the fine structure splitting in deuterium, and that of the hyperfine splitting in the ground state of atomic hydrogen. These two measurements differed by about 20 ppm. The fine structure measurement had been accepted on the basis of the theory of atomic spectra, which was believed to be more soundly established theoretically than the QED theory, which describes the hyperfine structure.

This experiment demonstrated convincingly that the hyperfine structure measurement was correct, and that the accepted value of α was wrong. This was a major victory for the proponents of QED. It was, moreover, a demonstration of the unity of physics.

**QED = quantum electrodynamics*

MOST physicists would agree that the fundamental physical constants play an important role in their science. After all, lists of them are enshrined in full multidigital glory in elementary textbooks and some are associated by name with eminent pioneers of the field. Few of us, however, have much understanding of where those long strings of numbers come from (found engraved on a stone tablet on a mountain, perhaps?) and fewer still ever have need for more than the first couple of digits and order of magnitude. The thought of actually measuring a fundamental constant to umpteen significant figures is likely to strike us like engraving the Constitution on the head of a pin-an amazing feat, but hardly very useful.

This is the story of some workers in the notoriously messy and imprecise field of low temperature solid state physics who, finding themselves in possession of a new tool for measuring a fundamental constant, forsook honest work to use it, with results that surprised everybody, including themselves. The story begins at the University of Pennsylvania in August, 1965. During the previous year I had been working together with Barry Taylor in an attempt to detect directly the microwave radiation emitted by the ac Josephson supercurrent in superconducting tunnel junctions. Barry had completed his Ph.D. thesis on electron tunneling (primarily in superconductors) at Penn under Eli Burstein in 1963 and had remained to continue this work. I had joined the Penn faculty in 1960. Earlier work with Barry and Eli, together with some experience with microwave electronics and a desire to learn something about superconductivity and the exciting new Josephson effects, had led me quite naturally into collaboration with Barry on the radiation detection experiment. In the spring of 1965 we had finally succeeded, with able assistance from a theorist, Doug Scalapino (one of that rare type who likes to twiddle knobs in the laboratory and can be prevailed upon to help move heavy equipment as well), and from a student of Barry's, Bob Eck (who would do anything to avoid writing the Ph.D. thesis he was supposed to be working on).

At the same time, Barry had accepted a position at the RCA Laboratories in Princeton. So in August we found ourselves in a borrowed office (hiding from importunate colleagues and the telephone) feverishly trying to complete a couple of papers on our experiments before Barry's departure for Princeton. Being no more enamored with writing than Mr. Eck, we frequently lapsed into discussions of



Donald Langenberg in his office at the University of Pennsylvania shortly after the measurement of e/h described in his Discovery Story. [Photo: Walter Holt].

various experiments suggested by our radiation experiment results. One of the ideas which emerged from our conversations was that perhaps somebody really ought to make a careful experimental test of the validity of the Josephson frequency-voltage relation, $\nu = 2eV/h$, which was supposed to relate the frequency ν of the ac Josephson supercurrent to the dc bias voltage across the junction. Why? There were several reasons, all rather naive in retrospect. I had been working in the field of 'Fermiology,' the study of Fermi surfaces in metals. Several years earlier there had arisen in that field the question of whether the effective electron charge in normal metals differs from the free electron charge, just as the effective electron mass differs from the free electron mass. The experimental answer appeared to be 'no' to a precision of about 1%, the same level to which the Josephson frequency-voltage relation had been checked in our own and in earlier ac Josephson effect experiments. The presence of the electron charge in the latter relation together with the relative simplicity of the required frequency and voltage

measurements suggested that one might be able to achieve a considerably more precise check on the possible charge deviation. Then there was the question of the '2' in the frequency-voltage relation. This had to do with bound superconducting pairs and all that, and did not seem exactly controversial at that point. Nevertheless, we thought it might be interesting to subject that number to a rigorous experimental check.

The idea that one might get something worthwhile out of the experiment in the form of a determination of e/h, even if no interesting deviations from the simple frequency-voltage relation turned up, followed quite rapidly and naturally. I remember looking up an article in the February, 1964, Physics Today, which gave the latest values of the constants, to see how well one would have to do to get a value comparable in accuracy with the accepted value. I was encouraged to find that the stated uncertainty in e/h was 29 ppm (parts per million). It did not seem impossible that one might be able to achieve an accuracy at least approaching that. (I did not at the time appreciate the import of the footnote stating that the quoted uncertainty represented a three-standard-deviation estimate, rather than the one-standard deviation I had in mind. This was fortunate, because an accuracy measured in single digit parts per million begins to sound a little scary, and we might have been far less optimistic about achieving such a goal. Ignorance is bliss!) So it looked as if one might win-however the experiment turned out. Either one would turn up an interesting and important deviation from the simple Josephson frequency-voltage relation, or one might be able to get a highly accurate determination of e/h. The latter might or might not contribute much to physics, but it would at least have the novelty of coming from solid state physics rather than the more elegant and precise fields of atomic and elementary particle physics from which fundamental constant determinations usually come.

Barry and I rapidly became so entranced with the idea of doing this experiment that we decided to proceed forthwith. It was clear that it would go better if Barry could remain at Penn rather than leave for RCA. Within a few days, we had arranged for a postponement of his arrival there and an extension of his faculty appointment at Penn, the latter within several weeks of the beginning of the fall term! (Oh, for the days when such things could be done without making a federal case of it!)

The next problem was manpower and equipment. It is well-known that in a university all the real research work is done by graduate students. It was clear that we needed a student working with

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us if we were to progress at a reasonable rate. So we called in Bill Parker, a student who had already been working with me for about a year and who had just passed his Ph.D. qualifying exam and was ready to start full-time research. I had originally suggested he begin work on something quite different, but now we presented our sales pitch and asked him if he'd like to have a go at the Josephson frequency-voltage relation. Both Barry and I remember having made a very hard sell to a rather dubious student-after all, measuring fundamental constants is what physicists did in the nineteenth century, and this was 1965. But perhaps it's best to let Bill describe this critical moment in his own words, taken from a letter he wrote me recently in connection with the preparation of this article:

> I first heard of the Josephson effect when, a day or so after passing my Ph.D. qualifying examination, you, Barry and Doug called me into a small office and made me an offer I couldn't refuse. You suggested that an investigation of the accuracy of the Josephson frequency-voltage relation was both an interesting experiment and a potential Ph. D. thesis. I was particularly intrigued by the suggestion that I would get a Ph. D. thesis regardless of how the experiment turned out; either any deviation from the simple Josephson frequency-voltage relation would be of intrinsic interest to the theory of superconductivity or a new determination of a fundamental constant would result. With very little hesitation and with no repressed doubts, I enthusiastically agreed to accept your offer to give up cyclotron resonance and join the effort to measure e/h. As it turned out, I have never made a better decision in my life except, of course, to marry my wife.

The equipment problem was not so quickly solved. The radiation experiment had been done with a lashup of borrowed and scrounged stuff, and if we were going into the business seriously we had to lay our hands on quite a lot of permanent equipment. Thus began a campaign for funds and donated equipment which lasted many months, but eventually brought in everything we needed.

Conceptually, the experiment we were undertaking was very simple. We had to bias a Josephson junction at a dc voltage V and measure the frequency of the resulting ac Josephson supercurrent, either by detecting the radiation it emitted, as in our earlier experiments, or by synchronizing it with an externally applied microwave signal and detecting the steps thereby induced in the dc current-voltage characteristic. (The latter effect had been discovered earlier by S. Shapiro.) The frequency and the voltage had to be measured as accurately as possible; the ratio would either yield 2e/h or an exciting surprise. Now the measurement of the frequency of a microwave signal with an accuracy as high as 0.01 ppm presents no particular

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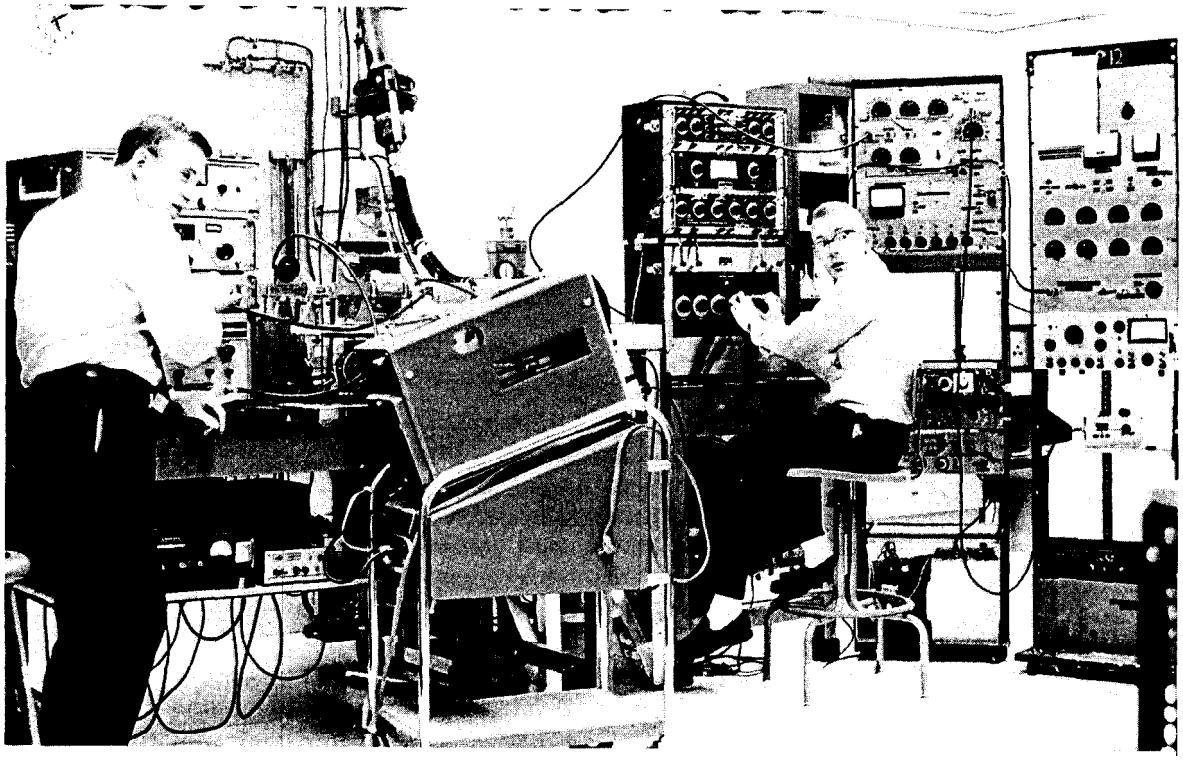
problem except the acquisition of the cash with which to buy the standard commercial equipment required. The crucial factor was measurement of the dc voltage. The voltages in question were small, several hundred microvolts at most, and the source was at cryogenic temperatures. Our main problem was to find out how to measure such a voltage accurately.

None of us knew very much about accurate dc voltage measurements. We'd all used commercial potentiometers and standard cells for measuring thermocouple potentials and the like, but we had no idea what the current state of the art was. Perhaps there was some marvelous, sophisticated, new method. We soon found out there wasn't: the null-balance principle and the electrochemical reference cell were still the basis for precision dc voltage measurements. There were, however, a variety of alternate circuits and methods, so we began a careful study of these options to see what the optimum system for the ultimate measurement might be.

While this proceeded, we began preliminary experiments using an old precision potentiometer Barry had unearthed at the Frankford Arsenal. This instrument had the polished wooden case, the form, and nearly the size of a coffin. It looked rather incongruous and archaic alongside our shiny new space-age electronics. However, we were rapidly learning that in precision dc instrumentation as in wine-making, 'well-made' and 'properly-aged' often imply 'good.' This instrument yielded a set of measurements with an uncertainty of 60 ppm, an advance in accuracy of two orders of magnitude. The result was a little lower than the accepted value of e/h, but the error bars overlapped comfortably, and we duly published it as a confirmation of the validity of the Josephson frequency-voltage relation to that accuracy. So far so good!

Meanwhile, we'd been consulting with every dc measurements expert we could find. We learned that while voltages of the order of magnitude of one volt could be measured with accuracies better than one ppm, the voltage division required to compare our few hundred microvolts with a standard cell near one volt would most likely increase the uncertainty to perhaps 10 ppm under the best of circumstances. While we were unenthusiastically contemplating getting into the potentiometer designing business, there appeared announcements from two different commercial firms of new potentiometers specially adapted to measurements of small voltages and with claimed accuracies of about 10 ppm. Providence and free enterprise had come to our rescue in the nick of time. Now our problem was reduced to that of deciding between the two and finding some more money. We soon did both, and placed an order.

All this brought us to the summer of 1966. Barry had been



Bill Parker (right) and Barry Taylor (left) in their laboratory at the University of Pennsylvania. The equipment shown was used in their sub-part-per-million accuracy determination of e/h. In the center of the photo is the dewar which held the Josephson junction, surrounded by the necessary electronic apparatus.

obliged to leave to take up his position at RCA in May. The new potentiometer arrived in July, and Bill began preparing for a series of serious measurements*. In August, I departed for a sabbatical leave in Paris, leaving the experiment in Bill's hands. By then it had become clear that Bill was an outstandingly capable student, so I could do so with few qualms. Besides, Barry was not far away and another student, Arnold Denenstein, who was to play an increasingly important role in our e/h program at Penn, was there to help with his unique skills. In any case, the subsequent scenario I envisioned involved a long and tedious acquisition of data accompanied by gradual discovery and elimination of innumerable sources of experimental error, yielding in perhaps a year or two a final value of e/h with an uncertainty of perhaps 10 ppm and in good agreement with the accepted value. I couldn't have been more wrong! Actual events moved much more rapidly and excitingly.

By the end of 1966 it was clear that our experimental e/h was *not* in agreement with the accepted value! Was the discrepancy due to solid state corrections to the Josephson frequency-voltage relation, was there a major undetected systematic experimental error, or was it possible that there was something wrong with the accepted official value of e/h? Bill had taken care to vary all the experimental parameters over wide ranges and had used a variety of different types of junctions and junction materials, in an effort to investigate the first possibility. Nothing seemed to affect the result. A great deal of thought and analysis turned up no source of systematic error of the required magnitude. That left the last possibility. It seemed incredible that the accepted value of a fundamental constant could he so

* This ultimately entailed pushing the accuracy of the instrument well beyond the manufacturer's claimed level through modifications and the design of special calibration procedures.

much in error, but a study of the most recent evaluation of the constants revealed an unresolved discrepancy. Everything suddenly fell into place! In Bill Parker's own words:

> One of the most exciting moments of my scientific career occurred one late evening in the winter of 1966 when, after hours of studying Cohen and DuMond's review, I suddenly realized that the Josephson value of e/h predicted a value of the fine structure constant α in agreement with the hyperfine value. As a consequence, one of the serious difficulties with quantum electrodynamics disappeared. At that moment I felt what must be the true excitement of scientific discovery.

The problem was that when Cohen and DuMond** had carried out their most recent evaluation of the constants, they were faced with two discrepant experimental values of the Sommerfeld fine-structure constant α. One came from measurements of the fine-structure splitting of atomic hydrogen, the other from measurements of the ground-state hyperfine splitting in atomic hydrogen. The theory of the latter included a somewhat controversial correction for the dynamic polarizability of the proton. Unable to resolve or to identify the source of the discrepancy, Cohen and DuMond had opted to use the result of the former experiment rather than the latter on the grounds that its theory was on much firmer ground. It turned out that a value of the fine-structure constant could readily be derived from our experimental value of e/h, and it indicated quite clearly that the fine-structure α was wrong and the hyperfine α was right! Because of the pivotal role played by by α in evaluations of the fundamental constants, this meant that many of the constants were significantly in error.

From this point on, Bill was more occupied in checking, rechecking, and refining the result, than in finding a job and writing his thesis. By the time I had thought he might just be beginning to zero in on an interesting and important but not enormously exciting result, he had passed his Ph.D. oral exam and was gone, amid an expanding wave of repercussions from this totally unexpected number.

** Cohen, E.R., and DuMond, J.W.M., *Rev. Mod. Phys.* 37, 537 (1965).

The impact of this result on us and on the physics community as a whole owed much to its unexpected source. An experiment in low temperature, solid state physics had yielded a result with important implications in such seemingly distant and more fundamental fields as atomic physics and quantum electrodynamics Bill recalls:

> being asked to discuss the e/h measurement before a solid state seminar. I was utterly amazed to discover that the large auditorium reserved for that seminar was filled by the majority of the particle physicists at Penn. Solid state people were in a minority at the first public discussion of this experiment in superconductivity.

One of the particle physicists is said to have remarked at the time that: "The most important elementary particle experiment in this department is being done by a graduate student in solid state physics!" Or, as an eminent particle theorist remarked much later upon being introduced to me: "Oh, you're the fellow who measured α in activated sludge!"

The (quite proper) skepticism of many that an important fundamental physical constant could be simply extracted from a complicated interacting many-particle, solid state system called for redoubled efforts to refine our result and to confirm that there really were no corrections we had overlooked. Hindsight lends credence to the argument that the superconducting state really does have the remarkable property of welding together 10^{23} electrons into a phase-coherent quantum state with all the simplicity (and more) of the one-electron quantum state of the hydrogen atom. But we had to be sure. Further experiments confirmed the earlier results and yielded a final value of e/h with an uncertainty of 2.4 ppm†. This was used as the basis for a thorough re-evaluation of the constants which provided revised values and identified several remaining discrepancies between experiment and theory in quantum electrodynamics‡. □

Philadelphia, August 1973

No QED Theory Involved

It should be noted that the input data in this experiment and the constants to be derived from them are independent of quantum electrodynamics (QED) only in the practical sense that it is not essential to make use of any QED theory in the analysis. In a deeper philosophical sense it is clear that these input data are dependent on electrodynamics interactions which may be described by some QED theory of the future.

† Parker, W.H., Langenberg, D.N., Denenstein, A., and Taylor, B.N., *Phys. Rev.* 177, 639 (1969).

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Said V. Hughes at a conference luncheon,
"The fine structure constant's a function
Of levels in muonium
And perhaps positronium,
But not any Josephson junction!"

S. L. KAUFMAN

Said D. Langenberg to V. Hughes,
"To each his particular views.
But so long as my dewar
Makes ppms fewer
I'll follow Josephson's clues."

D. N. LANGENBERG

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Experimental Test of the Josephson Frequency-Voltage Relation*†

D. N. LANGENBERG, W. H. PARKER, AND B. N. TAYLOR‡
Department of Physics and Laboratory for Research on the Structure of Matter, University of Pennsylvania, Philadelphia, Pennsylvania

(Received 31 May 1966)

The Josephson frequency-voltage relation in superconduction's $\nu = 2eV/h$ has been tested by measuring frequency-voltage ratios in Josephson junctions using three different Josephson-effect phenomena: (1) microwave emission from evaporated-film tunnel junctions, (2) microwave-induced quantum voltages in evaporated-film tunnel junctions, and (3) microwave-induced current steps in the *I-V* characteristics of evaporated-film tunnel junctions and of "weak-link" point-contact junctions. The use of the "harmonic effects" associated with each phenomenon to increase the accuracy of measurement, and the factors limiting the accuracy in each case, are discussed. The measured frequency-voltage ratios for the three methods were the same to within their respective experimental errors. The highest accuracy was obtained using microwave-induced current steps. The measured frequency-voltage ratio in this case was equal to the currently accepted value of $2e/h$ to within an experimental accuracy of 0.006% (60 ppm). The frequency-voltage ratio was also found to be independent of the type of junction used, temperature, magnetic field, harmonic number, voltage polarity, microwave power, and frequency to within the IO-ppm precision of the measurements.

THERE now exists a considerable body of experimental and theoretical information about phenomena which are direct consequences of long-range order and macroscopic quantization in super fluids. For superconductors, these phenomena include the quantization of magnetic flux and the dc and ac Josephson effects. (Analogous effects are also observed in superfluid helium.) The existence of these phenomena can be shown theoretically to follow quite simply and generally from assumptions of long-range order and macroscopic quantization, independent of any microscopic model of the superfluid.[1,2] In the case of the ac Josephson effect, the theory predicts the existence of an ac supercurrent between two weakly coupled superconductors maintained at a potential difference V, with a frequency ν given by the Josephson frequency-voltage relation $\nu = e^*V/h$, where h is Planck's constant and e^* is the charge of the "particle" which carries the supercurrent.[3] A connection with the microscopic theory is normally made by taking e^* to be $2e$, the charge of the Cooper pair. This choice has some experimental support: the quantum of magnetic flux has been shown to be equal to the theoretically predicted value $hc/2e$ to within 3%[4] and studies of the radiation emitted by the ac supercurrents in tunnel junctions have shown that the frequency-voltage ratio is $2e/h$ to an accuracy of about 1%.[5] However, experiments of the latter type have a potential accuracy considerably greater than 1%. Because of the fundamental part played by the Josephson frequency-voltage relation in the present general picture of superconductivity, we felt it was important to make full use of this potential to test the relation as rigorously as possible. In this paper, we report the results of such a test. We find that within our present experimental accuracy of 0.006% (60 ppm), the Josephson frequency-voltage ratio is indeed equal to $2e/h$, and that, within the 10 ppm precision of the measurements, the ratio is independent of all experimental parameters checked.

Three similar but operationally different experimental methods can be used to measure the Josephson frequency-voltage ratio; they are discussed in some detail below. In each case it is necessary to make an absolute measurement of a dc potential difference between two weakly coupled superconductors and a microwave frequency. In the present experiments the microwave frequency was in X band (8.0-12.4 GHz) and was easily measured to an accuracy of better than 10 ppm using a Hewlett-Packard frequency counter and microwave-transfer oscillator. The over-all accuracy of the results was limited solely by the accuracy of the voltage measurements. The factors determining this accuracy were the calibration of the potentiometer and standard cell used, the stability of the voltage to be measured, and the presence of spurious voltages in the measuring circuit (e.g., thermoelectric voltages). Such spurious voltages are usually a serious source of difficulty in accurate measurements of small dc voltages, but pose no serious problem in our experiments because of a unique property of Josephson junctions, the dc Josephson effect. This effect provides a zero reference voltage at the junction when the junction is operating in the

*A contribution from the Laboratory for Research on the Structure of Matter. University of Pennsylvania, covering research sponsored by the Advanced Research Projects Agency and the National Science Foundation.

† For a preliminary account of this work, see B. N. Taylor, D. N. Langenberg, and W. H. Parker, Bull. Am. Phys. Soc. 11, 191 (1966).

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superconducting dc Josephson state. (We have experimentally shown that the dc voltage across junctions biased in the dc Josephson mode is ≤ 5 nV, the limit of resolution of our present equipment; this was done by observing that no detectable change in galvanometer deflection occurred when the junction current was reversed. In addition, we have shown that the voltage across junctions biased in the dc Josephson mode is constant to within 5 nV for all values of current from zero up to the maximum dc Josephson current.) Any nonzero voltage then appearing in the measuring circuit must be spurious and can readily be corrected for, if it remains sufficiently constant during a measurement. In our experiments, the spurious voltages were typically 40 to 80 nV and remained constant to within 5 nV for the duration of several measurements. The voltage measurements were made using a calibrated Rubicon Model 2773 potentiometer with a Guildline photocell amplifier and galvanometer. The uncertainty due to the potentiometer system was of the order of ± 5 nV (determined by our ability to interpolate between the 10-nV steps of the last dial), or 60 ppm (determined by the accuracy of calibration of the instrument), whichever was larger. The precision of the voltage measurement was limited to about 10 ppm by the 5-nV resolution and linearity of the potentiometer, the stability of the standard cell, and the incidental frequency modulation on the microwave source, used to induce the effects discussed below. According to the Josephson frequency-voltage relation, the voltage corresponding to a frequency in X band is of order 20 μV ($2e/h$=484 MHz/μV), so that a measurement of a voltage directly related to an X-band frequency could be expected to have an accuracy of only 250 ppm. It was necessary, therefore, to use the "harmonic effects" associated with each of the three methods in order to make use of the full accuracy of the potentiometer.

Block diagrams illustrating the principal features of the three methods are shown in Fig. 1. The methods are:

A. Radiation Emission

Josephson tunnel junctions which displayed natural-mode structure in their *I-V* characteristics (i.e., current steps at essentially constant voltages) were biased at a high-order mode or step and the frequency of the radiation emitted in X band was measured.[5] Since this frequency was the nth subharmonic of the fundamental frequency corresponding to the bias voltage (cf. Ref. 5), the quantity derived from the frequency and voltage measurements was actually nv/V. (The modified Josephson frequency-voltage relation corresponding to this situation is $v = 2eV/nh$, where V is the bias voltage at the nth mode and v is the frequency of the detected radiation.) Typical values for n in these experiments ranged between 1 and 9. Radiation was observed with the junction biased at modes with $n > 9$, but the frequency of this radiation had a voltage dependence which

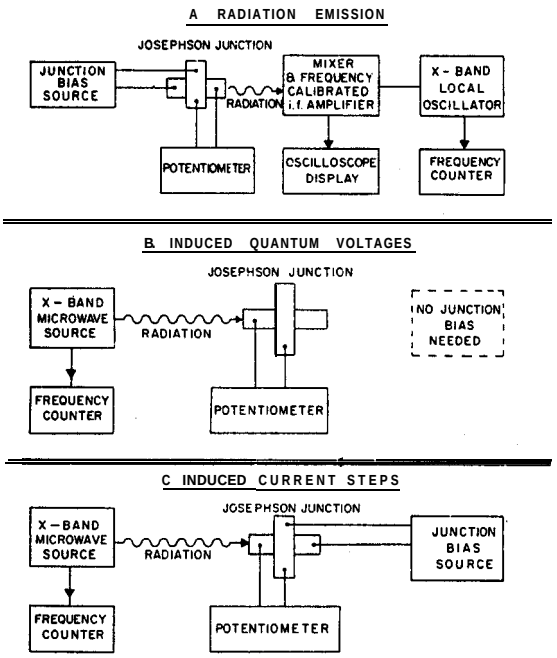


FIG. 1. Schematic block diagrams showing the three different methods used to test the Josephson frequency-voltage relation.

differed markedly from that of the Josephson radiation. (The properties of this "non-Josephson" radiation have been studied and will be reported elsewhere.[6]) This effect limited the maximum voltage to be measured to approximately 200 μV. This method has therefore not yielded as accurate a test of the Josephson frequency-voltage relation as the third method described below.

B. Induced Quantum Voltages

Josephson tunnel junctions were irradiated with X-band microwaves and the dc quantum voltages induced across the junctions were measured.[7] The equation relating the frequency of the applied radiation to the induced quantum voltages is again $v = 2eV/nh$. In this case, n corresponds to the quantum number of the induced voltage V, and v is the frequency of the applied radiation. In these experiments, values for n ranged between 1 and 11. The voltages corresponding to the larger values of n were rather unstable, i.e., the junctions tended to switch spontaneously from one quantum voltage to another. For this reason, the larger voltages were quite difficult to measure and this caused a corresponding increase in uncertainty.

C. Induced Current Steps

Josephson junctions were irradiated with X-band microwaves and steps in the current at discrete voltages

[6] D. N. Langenberg, W. H. Parker, and B. N. Taylor, Phys. Letters 22, 259 (1966).

[7] For a discussion of these quantum voltages, see D. N. Langenberg, D. J. Scalapino, B. N. Taylor, and R. E. Eck, Phys. Letters 20, 563 (1966).

were induced in the junction *I-V* curve as originally observed by Shapiro.[8] These induced steps were more stable than the induced quantum voltages, so that more accurate voltage measurements could be made. The steps were observed to be vertical (i.e., the voltage at each step was constant over the full range of current on the step) to within the 5-nV resolution of our equipment. The equation $v = 2eV/nh$ again gives the relation between the frequency of the applied microwaves and the voltage at the nth current step. In our experiments, values of n from 1 to 88 were used (note that for the larger n values, V was of order 2 *mV*). With this method, we were able to check whether the results depended on details of the experimental situation. All of the available experimental parameters were varied and several types of junctions were used. Data was obtained for a large number of different n values, for both voltage polarities, for values of magnetic field between 0 and 10 G, for $0.3 < T/T_c < 0.9$ (on an evaporated-film tin-tin-oxide-tin junction), for a 10-db variation of incident microwave power (approximately 0.1 to 1 mW), and for a frequency range of about 9 to 12 GHz. Data were obtained from a number of evaporated thin-film tin-tin-oxide-tin junctions, from an evaporated tin-tin-oxide-lead junction which contained a superconducting bridge, and from niobium-tantalum "weak-link" point-contact junctions.[9] Since stable current steps could be induced at very high values of n, this method yielded results with an accuracy limited only by the uncertainties in the calibration of the potentiometer system (60 ppm). No dependence of the results on any of the experimental parameters or junction types was found to the 10-ppm precision of the measurements.

Results obtained using all three methods are summarized in Table I and compared with the current "best" value of $2e/h$.[10] We conclude that, to within the uncertainties stated in the table, the Josephson frequency-voltage ratio is equal to $2e/h$ in several different types of junctions and under a wide variety of experimental conditions. It should be reiterated that the stated uncertainty of 60 ppm for the third method (induced current steps) is attributable entirely to calibration uncertainty in our present voltage-measuring system and not to any fundamental experimental difficulties. It appears that this method (and also the others, with improvements in experimental technique) can be used to achieve a direct determination of e/h with an accuracy significantly better than has been achieved using other methods. We are currently engaged in making this determination.

TABLE I. Comparison of the Josephson frequency-voltage ratios obtamed m the present work with 2 times the valu e of e/h derived by Cohen and DuMond in their 1963 least-squares adjustment of the fundamental constants (Ref. 10). The experimental values have been corrected for the difference between the absolute volt and the legal volt as maintained by the National Bureau of Standards (1 NBS volt= 1.000012 ± 0.000004 absolute volt). All of the uncertainties indicated represent approximately a 70% confidence level.

Method	Josephson frequency-voltage ratio, in MHz/μV
A. Radiation emission	483.62±0.05
B. Induced quantum voltages	483.59±0.15
C. Induced current steps	483.59±0.03
2 (e/h) = 483.610 ± 0.005 MHz/μV	

We should like to acknowledge valuable conversations with R. P. Feynman, J. E. Mercereau, and D. J. Scalapino, and to thank A. Denenstein for his technical assistance in these experiments.

[8] S. Shapiro, Phys. Rev. Letters 11, 80 (1963); S. Shapiro, A. R. Janus, and S. Holly, Rev. Mod. Phys. 36, 223 (1964).

[9] It has been shown by several workers that weak links can exhibit properties very similar to those of Josephson tunnel junctions. See, for example, P. W. Anderson and A. H. Dayem, Phys. Rev. Letters 13, 195 (1964); J. E. Zimmerman and A. H. Silver, Phys. Rev. 141, 367 (1966).

[10] E. R. Cohen and J. W. M. DuMond, Rev. Mod. Phys. 37, 537 (1965).

MEASUREMENT OF *2e/h* USING THE ac JOSEPHSON EFFECT AND ITS IMPLICATIONS FOR QUANTUM ELECTRODYNAMICS*

W. H. Parker
Department of Physics and Laboratory for Research on the Structure of Matter, University of Pennsylvania, Philadelphia, Pennsylvania

and

B. N. **Taylor†**
RCA Laboratories, Princeton, New Jersey

and

D. N. Langenberg
Department of Physics and Laboratory for Research on the Structure of Matter, University of Pennsylvania, Philadelphia, Pennsylvania

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Using the ac Josephson effect, we have determined that 2e/h = 483.5912 ±0.0030 MHz/μV. The implications of this measurement for quantum electrodynamics are discussed as well as its effect on our knowledge of the fundamental physical constants.

In this Letter, we report a high-accuracy measurement of *2e/h* using the ac Josephson effect (here, *e* is the electron charge and *h* is Planck's constant). When combined with the measured values of other fundamental constants, this measurement yields a new value for the fine-structure constant α which differs by 21 ppm from the presently accepted value. This change in a removes the present discrepancy between the theoretical and experimental values of the hyperfine splitting in the ground state of atomic hydrogen, one of the major unsolved problems of quantum electrodynamics today. We also discuss the effect of this change on our present knowledge of the fundamental physical constants.

The phenomenon used in these experiments was first predicted by Josephson in 1962.[1] He showed theoretically that when two weakly coupled superconductors are maintained at a potential difference *V,* an ac supercurrent of frequency

$$\nu = (2e/h)V \qquad (1)$$

flows between them. This equation, known as the Josephson frequency-voltage relation, can be shown to follow from quite general assumptions concerning superconductivity,[1,2] and is believed to be exact. In a recent publication[3] we reported an experimental test of this relation which verified that the frequency-voltage ratio was equal to the then best value of $2e/h$ to within the 60-ppm uncertainty of the measurements, and was completely independent of all of the experimental variables tested. This work also demonstrated that the main experimental difficulty with using the ac Josephson effect to make a high-accuracy determination of $2e/h$ was the calibration of the voltage-measuring system. Since then, we have acquired a completely self-calibrating system, and have carried out a new series of measurements with an order-of-magnitude improvement in accuracy.

In these new experiments, measurements were performed on two types of "junctions" which show ac Josephson-effect phenomena-evaporated thin-film tunnel junctions, and point-contact weak links.[3] The particular phenomenon used was that of microwave-induced, constant-voltage current steps[1,2] of the type first observed by Shapiro.[4] Such steps appear as excess dc currents in the current-voltage characteristics of Josephson junctions when they are irradiated with microwave radiation of frequency ν. Physically, this effect arises from the presence of a microwave-induced ac voltage across the junction which frequency-modulates the ac Josephson current. The constant-voltage current steps simply correspond to zero-frequency (dc) sidebands. The relationship between the voltages V_n at which these steps occur and the frequency of the applied radiation is $2eV_n = nh\nu$, where n is the number of the step. A determination of $2e/h$ can thus be made simply by measuring the frequency of the applied microwave radiation and the absolute voltage at which the current steps occur. No other measurements are required except those necessary in calibrating the equipment, and in this sense it is a remarkably straightforward fundamental constant experiment.

The applied microwave radiation was generated by an X-band (8- to 12.4-GHz) oscillator with a stability of one part in 10^8 per hour when phase-locked to a quartz crystal reference. The frequency of the radiation was measured to an accuracy of one part in 10^8 by using an electronic counter and a microwave frequency converter. The reference time base of the counter was maintained to an accuracy of better than one part in 10^8 by regular phase comparisons with the U. S. frequency standard as broadcast by radio station WWVB, Fort Collins, Colorado. Thus, the frequency measurement contributed negligible error (~0.01 ppm) to the measured value of $2e/h$. The over-all accuracy was limited by the voltage-measuring system, i.e., potentiometer and standard reference voltage.

The reference voltage used was the mean voltage of a set of six standard cells, calibrated by the U. S. National Bureau of Standards (NBS), in a constant-temperature air bath. This voltage was known to 1 ppm in terms of the NBS legal volt, the uncertainty being an estimate of the possible changes in the emf of the standard cells due to transporting them from NBS to our laboratory. In order to obtain a value of $2e/h$ in absolute units, it is necessary to convert from the legal volt to absolute volts. The present best value of this conversion factor is 1 NBS legal volt = 1.000 012 ± 0.000 004 absolute volts, where the 4-ppm uncertainty is intended to represent a 50% confidence level.[5]

The potentiometer used was the Julie Research Laboratories PVP 1001.[6] This nanovolt instrument is self-calibrating in that it has provisions which enable the operator to measure all factors which contribute to the accuracy of a voltage measurement and to make any necessary corrections. Using techniques developed by Julie Research Laboratories[7] and NBS,[8] the 1-mV full-scale range can be calibrated with an rms uncertainty of between 3 and 4 ppm (this was the range normally used since the voltages of the induced current steps rarely exceeded 1 mV). The null detector used with the potentiometer consisted of a photocell amplifier and galvanometer and had a resolution of 1 nV.

In making accurate measurements of such small voltages, great effort is necessary to eliminate or correct for spurious voltages in the measuring circuit. In the measurements described here, the effect of voltages which do not reverse when the current is reversed (thermoelectric voltages, for example) was eliminated by measuring constant-voltage current steps of both polarities. Voltages which

reverse with current (those from Ohmic sources, for example) were shown to be negligible by observing that the voltage in the measuring circuit was constant over the full range of the zero-voltage current arising from the dc Josephson effect.[1] Spurious voltages due to any rectification of the microwaves were also shown to be negligible by observing that the measured value of *2e/h* was independent of microwave power over a range of 10 dB for a given sample and of 20 dB from sample to sample.

The results of measurments on several thin-film tunnel junctions and point-contact weak links are given in Table I. The standard deviation of a set of measurements obtained during one run[9] on any particular junction, typically 2 ppm, is due to the 1-nV resolution of the null detector, the stability and linearity of the potentiometer, and the stability of the thermoelectric voltages in the measuring circuit (usually of order 100 nV). Within this 2-ppm standard deviation, the measured value of *2e/h* was found to be independent of a wide variety of experimental conditions, including step number up to $n = 40$, magnetic field from 0 to 10 G, microwave frequency from 8 to 12 GHz, and microwave power. All of the measurements were carried out between 1.2 and 1.6°K.

For most of the current steps from which the data were obtained, the voltage was constant to within 1 nV (the resolution of the null detector) over the full range of the step. However, for three of the point contacts (marked by an asterisk in Table I), the voltage was found to increase by 5-10 nV as the current was increased over the range of the step. It was observed that in higher resistance point contacts (several tenths of an ohm rather than several hundredths of an ohm), where the voltage variation was as much as 200 nV, the midcurrent point of the step gave a value of *2e/h* equal to the average of all the data obtained on the constant-voltage steps. As the resistance was decreased, the voltage variation decreased and the midpoint continued to-give a value of *2e/h* in agreement with the constant-voltage step data. Extrapolating this behavior to contacts with 5- to 10-nV variation, we assume that the midcurrent point corresponds to the voltage at which the step would occur if it were constant. Thus, all measurements on such steps were made at this midpoint.

The average of all of the data in Table I, weighted as the inverse square of the rms uncertainties, gives

$$2e/h = 483.5912 \pm 0.6030 \text{ MHz}/\mu\text{V},$$

or in more conventional terms,

$$h/e = 4.135\ 725 \pm 0.000\ 026 \times 10^{-15} \text{ J sec/C}$$

$$= 1.379\ 529 \pm 0.000008 \times 10^{-17} \text{ erg sec/esu}.$$

The quoted uncertainty, about 6 ppm (70% confidence level), is an rms sum of all known sources of error, either systematic or random, and includes the uncertainty in the calibration of the potentiometer, the standard deviation

Table I. Summary of experimental data. Junctions of the form Sn-SnO-X are evaporated thin-film tunnel junctions while the others are point-contact weak links. The table entries are in chronological order and the decreasing uncertainty in the potentiometer calibration results from improved techniques.

JUNCTION	UNCORRECTED nν/V WITH STD. CELL. MHz/μV	POTENTIOMETER CORRECTION WITH rms UNCERTAINTY ppm	STD. CELL TEMPERATURE CORRECTION ppm	CORRECTED nν/V WITH rms UNCERTAINTY MHz/μV
Sn-SnO-Sn	483.610 ± .002	-30 ± 10		403.596 ± .005
Sn-Sn	483.6126 ± .0011	-2s ± 7		483.5986 ± .0032
Ta-Ta	483.6164 ± .0009	-3s ± 4	-0.5	483.5973 ± .0021
Sn-SnO-Sn	483.6156 ± .0008	-3s ± 4	-0.5	483.5965 ± .0021
Sn-SnO-Sn	483.6158 ± .0007	-4.2 ± 4	-0.3	483.5955 ± .0021
Sn-SnO-Pb	483. 6174 ± .0013	-43 ± 4	- 1.0	483.5962 ± .0023
Nb-Ta*	483. 6195 ± .0007	-42 ± 4	-1.0	483.5987 ± .0021
Ta-Nb_3Sn*	483 .6185 ± .0011	-44 ± 4	-0.5	483.5975 ± .0022
Ta-Ta*	483 .6185 ± .0016	-46 ± 4	-0.4	463.5961 ± .0025
Sn-SnO-Sn	483.6194 ± .0005	-45 ± 4	-0.0	483.5976 ± .0020
WEIGHTED AVERAGE OF DATA IN TERMS OF NBS VOLT MHz/μV_{NBS}				483.5971 ± .0022
NBS VOLT TO ABSOLUTE VOLT CONVERSION				- 0.0058 ± .0019
FINAL VALUE FOR 2e/h IN ABSOLUTE UNITS MHz/μV				483. 5912 ± .0030

of a set of measurements, the uncertainty in the absolute value of the NBS legal. volt, and the uncertainty in transferring the NBS volt to our laboratoratory. It should be noted that the standard deviation of the eight most accurate measurements, made over a period of several months, is only 2 ppm, an indication of the high precision of the measurements. To within this 2-ppm precision, the measured value of *2e/h* is independent of the material and type of junction used.

A value of the fine-structure constant can be derived from our value of *2e/h* and other directly measured quantities by use of the equation

$$\alpha^{-1} = \left[\frac{c}{4R_\infty}\frac{\mu_p}{\gamma_p}\frac{2e}{\mu_0 h}\right]^{1/2}, \qquad (2)$$

where c is the velocity of light, R_∞ is the Rydberg constant for infinite mass, γ_p is the gyromagnetic ratio of the proton, and μ_p/μ_0 is the magnetic moment of the proton in units of the Bohr magneton. Taking the best values for these quantities,[10] $c = 2.997925 \times 10^8$ m sec^{-1} ± 0.3 ppm, $R_\infty = 1.097\,373\,1 \times 10^7$ m$^{-1} \pm 0.1$ ppm, $\mu_p/\mu_0 = 1.521\,032\,5 \times 10^{-3} \pm 0.5$ ppm, $\gamma_p = 2.675\,192 \times 10^8$ rad sec^{-1} T$^{-1} \pm 3$ ppm, and $2e/h = 4.835\,912 \times 10^{14}$ Hz V$^{-1} \pm 5$ ppm, Eq. (2) gives[11]

$$(\alpha^{-1})_{2e/h} = 137.0359 \pm 0.0004.$$

This value is 21 ± 5 ppm less than the presently accepted value derived from. the fine-structure splitting (fs) in deuterium as measured by Treibwasser, Dayhoff, and Lamb,[12,13]

$$(\alpha^{-1})_{fs} = 137.0388 \pm 0.0006.$$

The new value of α derived here entirely removes the apparent discrepancy between the theoretical and experimental values for the hyperfine splitting (hfs) in the ground state of atomic hydrogen. This splitting (ν_{hfs}) has been measured to the extraordinary accuracy of 2 parts in 10^{11} by Crampton, Kleppner and Ramsey.[14] The quantum-electrodynamic expression for the splitting, which includes all theoretical effects other than the dynamic polarizability of the proton, is believed to be accurate to a few ppm.[15] When this expression is evaluated using $(\alpha^{-1})_{fs}$, it predicts

$$\frac{\nu_{hfs}(\text{expt}) - \nu_{hfs}(\text{theory})}{\nu_{hfs}(\text{expt})} = 43 \pm 12 \text{ ppm}.$$

[The quoted errors include an uncertainty of 2 ppm in the estimate of form factors and 5 ppm in $(\alpha^{-1})_{fs}$.] If the theoretical expression is evaluated using $(\alpha^{-1})_{2e/h}$, it predicts

$$\frac{\nu_{hfs}(\text{expt}) - \nu_{hfs}(\text{theory})}{\nu_{hfs}(\text{expt})} = 0 \pm 8 \text{ ppm}.$$

Although at present it is impossible to calculate the proton polarizability exactly, the best estimates indicate that it would increase ν_{hfs}(theory) by less than 10 ppm.[15] Thus, unless the proton polarizability is much larger than is presently believed, $(\alpha^{-1})_{fs}$ suggests a breakdown of quantum electrodynamics, while $(\alpha^{-1})_{2e/h}$ is consistent with both quantum electrodynamics and a small proton polarizability.

The change in α implied here is also important because in the 1963 adjustment of the fundamental physical constants by Cohen and DuMond, the value $(\alpha^{-1})_{fs}$ was used as an input datum. Because of the pivotal role played by α in this adjustment,"' any change in α will cause large changes in the values of the other fundamental constants. In Table II, we give the values of some of the more important constants which would have resulted if $(\alpha^{-1})_{2e/h}$ had been used as an input datum in the 1963 adjustment.

We might also point out that with this new experimental value for *2e/h,* a new and more reliable value for the x-ray wavelength conversion factor Λ can be obtained. Using the recent experimental data of Spijkerman and Bearden[17] for the voltage-to-wavelength conversion factor, $V\lambda_S$, we find that $\Lambda = 1.002\,067 \pm 0.000\,023$ Å/kxu based on Bearden's new definition of the x unit.[18]

Although we believe that the results reported here are highly reliable, work is continuing to ensure that there is no unknown systematic error in the measurements. To this end, experiments of the type described here are being carried out at higher frequencies (-70 GHz) as well as experiments involving the measurement of the frequency of the radiation emitted by a Josephson tunnel junction when biased to a known voltage. Preliminary results from experiments of the latter type are in complete agreement with the results presented here.

We should like to thank Mr. Loebe Julie for helpful discussions concerning the potentiometer, Mr. A. G. McNish for arranging the cal-

Table II. Changes in some of the fundamental physical constants resulting from substituting $(\alpha^{-1})_{2e/h}$ for $(\alpha^{-1})_{fs}$ in the input data of the 1963 adjustment. (N is Avogadro's number and m is the rest mass of the electron.) The numbers in parentheses are the one-standard-deviation errors in ppm.

Quantity	Units	Value given by 1963 adjustment	Value implied by this measurement	Change (ppm)
α^{-1}	—	137.0388(4)	137.0359(3)	−21
e	10^{-19} C	1.602 10(13)	1.602 20(13)	+63
	10^{-10} esu	4.802 98(13)	4.803 28(13)	
h	10^{-34} J sec	6.625 59(24)	6.626 28(24)	+105
m_e	10^{-31} kg	9.109 08(14)	9.109 65(14)	+63
N	10^{26} kmole^{-1}	6.022 52(15)	6.022 14(15)	−63

ibration of the standard cells, Dr. E. Richard Cohen and Dr. Jesse W. M. DuMond for their interest and encouragement, Professor S. A. Bludman and Professor D. J. Scalapino for helpful discussions, and Mr. A. Denestein for his excellent technical assistance.

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†Part of this work was performed while the author was at the University of Pennsylvania and part while at RCA Laboratories.

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[3]D. N. Langenberg, W. H. Parker, and B. N. Taylor, Phys. Rev. 150, 186 (1966).

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[8]R. F. Dziuba and T. M. Souders, "A Method for Calibrating Volt Boxes with Analysis of Volt-Box Self-Heating Characteristics", IEEE International Convention Record, March 1966, Pt. 10.

[9]A typical run might consist of 40 separate measurements on about 8 different steps.

[10]E. R. Cohen and J. W. M. DuMond, Rev. Mod. Phys. 37, 537 (1965).

[11]Throughout this Letter, all uncertainties correspond to one standard deviation unless otherwise stated. The uncertainties for γ_p and $2e/h$ do not include the uncertainty in the conversion factor from NBS "as maintained" electrical units to absolute units because the conversion factor for each of these quantities is essentially the same and enters Eq. (2) in such a way as to cancel out. Thus, we are able to completely bypass the problem of the relationship between NBS legal units and absolute units.

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[15]For an up-to-date review of the theoretical situation and the original references, see S. D. Drell and J. D. Sullivan, Phys. Rev. (to be published).

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SUBSEQUENT DEVELOPMENTS

1. After some of the dust created by the Parker-Taylor-Langenberg experiment had settled, we returned to the problem of pushing the ac Josephson effect determination of e/h to the limit of its potential. This involved the design and construction of two voltage comparators specially adapted to the experiment. Using these, we carried out a determination with a precision of 0.03 ppm and an accuracy of 0.12 ppm.* This further increase in accuracy of a factor of 20 was limited by uncertainties associated with the definition and maintenance of the reference volt itself, and therefore is about as far as it's possible to go without fundamental changes in the way the reference volt is handled.

* Finnegan, T.F., Denenstein, A., Langenberg, D.N., *Phys. Rev.* B **4,** 1487 (1971). See p. opp.

2. The obvious applicability of the Josephson junction as an 'atomic' voltage standard is currently being exploited in many national standard laboratories. Beginning July 1, 1972, the United States National Bureau of Standards has maintained the U.S. legal volt using the ac Josephson effect and the voltage comparators mentioned above. This changeover was carried out under the leadership of Barry Taylor, now Chief of the Absolute Electrical Measurements Section of the Electricity Division of the NBS.

3. The question of the validity of the Josephson frequency-voltage relation has attracted a great deal of theoretical and experimental attention. The relation has withstood all assaults to date. It's independence of a variety of parameters has been demonstrated experimentally to the 0.01 ppm level, and no corrections have been convincingly indicated theoretically at a level exceeding about 10^{-10}.

D. Langenberg

LATEST RESULT

In 1970, the new and improved measurement of Finnegan, Denenstein and Langenberg led to a determination of *2e/h* with an accuracy approximately 20 times greater than the first measurement: it was determlned to an accuracy of *0.12 part in a million!* The final result is given as:

$$2e/h = (4.83593718 \pm 0.0000006) \times 10^{14}\ \mathrm{Hz/V_{NBS}}$$

(V_{NBS} is the National Bureau of Standards as-maintained standard volt).

The intrinsic precision of this measurement establishes a basis for a practical Josephson voltage standard with significant advantages over a standard based on an electrochemical cell.

This measurement yields a value for the Fine Structure Constant of:

$$\frac{1}{\alpha} = 137.03602 \pm 0.00021 \ (1.5 \text{ ppm}).$$

The uncertainty in α is now completely dominated by the uncertainty in the gyromagnetic ratio of the proton γ_p.

IMPORTANT REMARKS FOR EXPRIMENTALISTS*

B. N. Taylor, W. H. Parker, and D. N. Langenberg

Recommendations for Reporting Experimental Results

During the preparation of this review, it became apparent that our task would have been much easier if certain procedures had been generally followed in the reporting of experimental results. More important, we would have been able to compare theory with experiment and to incorporate data into our least-squares adjustment with a great deal more confidence. We would therefore like to make several suggestions for presenting experimental results which, if followed, will facilitate the work of future reviewers and materially increase the reliability of direct comparisons between theory and experiment and the output values of any least-squares adjustment.

(1) It should be made quite clear exactly what steps were taken to arrive at the final result from the actual experimental observations. If at all possible, the data analysis should be presented in such a way that the reader can follow each important step of the analysis and duplicate the final reported results. This means that particular attention must be paid to the way in which presumably well-known constants and corrections of various kinds enter the calculations. The experimenter should ask himself, *"Have I provided enough information in a sufficiently clear manner so that 10 years hence my result can be updated in the light of any new information or data which may become available?"*

(2) Every effort should be made to discussin some detail each important systematic error believed to be present in the experiment and to list estimates of these errors in tabular form, preferably as 70% confidence-level estimates (i.e., standard deviations). We strongly deplore statements such as "the systematic error is not believed to exceed 0.5" without any further discussion of the individual systematic errors contributing to this value or whether the quoted value is meant to be a 50%, 70%, or 95% confidence-level estimate. We also urge that the statistical uncertainty be computed using standard methods and presented in the form of a standard deviation. Since the uncertainty assigned a particular experimental datum plays a major role in determining whether it is in agreement or disagreement with theory and how much weight it will carry in a least-squares adjustment, correct error estimates are of the utmost importance.

(3) The numerical value of a datum and its uncertainty should be presented with a sufficient number of significant figures to prevent rounding errors when it is used. As a rule of thumb, the error should usually be stated to two-digit accuracy, and the datum itself should contain enough digits so that the quoted uncertainty corresponds to the last two figures.

*Reprinted from the *Reviews of Modern Physics*, Volume 41, Number 3, July 1969.

(4) When the experiment is completed and the results published, the original laboratory notebooks and other pertinent information should be deposited in duplicate in secure places; use of a departmental safe or bank vault is not presumptuous, but prudent. No matter how detailed the published report may be, these items constitute the *only* really complete record of the results of the expenditure of man-years of effort and many thousands of dollars of someone's money. Their preservation may prevent much uncertainty and unnecessary future expenditures.

We recognize that these recommendations will be considered a nuisance by some workers, but it seems to us that anyone who can spend two or three (or perhaps five) years measuring a particular quantity can certainly spend an extra month or two writing up his results in a manner which will maximize their future usefulness. We also believe rather strongly that the responsibilities of an experimenter include assuring complete and open access to his results and procedures for a considerable time after the completion of his experiments.

Lest our theoretical colleagues feel slighted, may we suggest that when they write up the results of their calculations, they do the following: (1) Clearly summarize the final results in one place so that the reader doesn't have to search through innumerable pages. (2) Discuss the effects of the important approximations and/or those of uncalculated terms. (3) Present the results in such a way that corrections can easily be made at a later date for changes in any of the numerical constants used in the calculations. (4) Discuss any differences, even if they are apparently insignificant, between their results and those of other workers. (5) Keep in mind that experimentalists may wish to read and understand the paper and use its results.

EXPLANATORY PHYSICS NOTES FOR NON-SPECIALISTS

PHENOMENOLOGICAL FOUNDATIONS OF THE JOSEPHSON EFFECT

Coupled Superconductors

B. D. JOSEPHSON

Cavendish Laboratory, Cambridge, England

I. INTRODUCTION AND BASIC FORMULAS

In this paper, I wish to describe the clarification which has taken place in our ideas on the nature and behavior of tunneling supercurrents since the original work was done on the subject, and to mention some recent experimental work. Tunneling supercurrents originally emerged out of a complicated calculation of tunneling currents as terms which did not go to zero when the voltage across the barrier was put equal to zero.[1] It is now clear, however, that they are of essentially the same nature as the familiar kind of supercurrent. The link between the two is provided by the Ginzburg-Landau theory,[2] and I shall first give a brief summary of this theory.

The first attempts to describe the superconducting phase transition, such as the Gorter-Casimir two-fluid model,[3] used a real order parameter, the density of superconducting electrons. These theories are based on an assumed dependence of the free energy on the order parameter, whose value is found by minimizing the free energy with respect to it. This type of theory explains the existence of a critical magnetic field and the second order phase transition in zero field, but cannot, without additional assumptions, attempt to give a description of normal-superconducting phase boundaries or the penetration region in a magnetic field. The Ginzburg-Landau theory describes these in a natural way. Unlike the two-fluid model it uses a complex order parameter ψ, interpreted as the wave function of the superconducting electrons. Thus $|\psi|^2$ takes the place of the density of superconducting electrons in the two-fluid model. The other new feature is a term in the free energy representing the kinetic energy of the superconducting electrons; this is where the complex nature of ψ enters. The kinetic energy term involves both ψ and the electromagnetic vector potential, and therefore couples together the electrons and the magnetic field. It thus turns out to be a natural consequence of the Ginzburg-Landau theory that the thermodynamically stable state of a superconductor in a magnetic field is a current carrying one. Since the Ginzburg-Landau theory, microscopic theories of superconductivity have been developed, giving an explanation of its assumptions in so far as they apply; but as they do not add anything essentially new to the main discussion we shall ignore them except in Sec. V.

After these preliminaries, let us return to tunneling between superconductors. In the first place let us consider a system consisting of two superconductors completely isolated from each other. Like the Schrodinger equation, the Ginzburg-Landau equations have the feature that from any given solution others may be obtained by changing the phase of ψ. The complete isolation of our superconductors from each other implies that it must be possible to alter the phases of ψ in each independently. On the other hand in a single (simply connected) superconductor under given external conditions ψ is completely determined apart from a phase factor, so that all phase differences are fixed. If now we imagine the two superconductors to be separated by a barrier of normal metal or insulator whose thickness is gradually reduced to zero, it is reasonable to suppose that the properties of the system go over continuously from those of two isolated superconductors to those of a single superconductor. This must happen in the following way: the free energy of the system must contain a term which depends on the relative phases of the values of ψ on the two sides of the barrier and increases in magnitude as the barrier becomes thinner. When the barrier is thick the coupling energy is very small and the phases are free to change arbitrarily, but once the coupling energy becomes large compared with kT, the phases become effectively locked together. We assume the coupling free energy to be expressible as an integral over the barrier:

$$F = \int_S f(P)dS\,, \tag{1.1}$$

where $f(P)$ depends only on the nature of the barrier and the local values of ψ on the two sides of it (and possibly on temperature). Throughout this paper we

[1] B. D. Josephson, Phys. Letters 1, 251 (1962). Owing to a computational error the value of j_1 given in this paper is incorrect. For the correct value see Ref. 24.

[2] V. L. Gineburg and L. D. Landau, Zh. Eksperim. i Teor. Fiz. 20, 1064 (1950).

[3] See, for example, D. Shoenberg, *Superconductivity* (Cambridge University Press, 1960), Chaps. 3 and 6.

[4] L. P. Gor'kov, Zh. Eksperim. i Teor. Fiz. 34, 735 (1958), [English transl.: Soviet Phys.—JETP 7, 505 (1958)] and Zh. Eksperim. i Teor. Fiz. 36, 1918 (1959) [English transl.: Soviet Phys.—JETP 9, 1364 (1959)].

shall be concerned only with small fields and currents, which do not appreciably affect the density of superconducting electrons $|\psi|^2$, and so it will be sufficient to regard the surface free energy as a function of ϕ, the difference between the phases of ψ on the two sides of the barrier.[6]

Associated with the surface coupling energy is the possibility of supercurrents through the barrier. In fact the two are intimately related:

$$j = \frac{2e}{\hbar} \frac{df}{\partial \phi} , \tag{1.2}$$

where j is the supercurrent through unit area of barrier and Gaussian units are used. Equation (1.2) is reminiscent of some formulas of classical thermodynamics, and can indeed be proved by classical thermodynamical means.[6] Anderson[7] has given a quantum mechanical derivation. Owing to the definition of ϕ as a phase difference, j and f must clearly be periodic functions of ϕ. The microscopic theory[1] shows that, in the limit of weak coupling, only first harmonics enter:

$$j = j_1 \sin \phi , \tag{1.3}$$

$$f = -\frac{\hbar}{2e} j_1 \cos \phi + \text{constant} . \tag{1.4}$$

(1.3) and (1.4) are good approximations for the barriers normally used in tunneling experiments.

It is sometimes convenient to use a gauge in which there is a large component of vector potential in the barrier normal to it; in this case, a gauge term,

$$-(2e/\hbar c) \int_1^2 \mathbf{A} \cdot d\mathbf{s}$$

must be added to ϕ, the integral being along a curve joining the superconductors?

II. EQUILIBRIUM PROPERTIES

In this section, we shall assume that the coupling across the barrier is sufficiently strong to maintain a definite phase relationship between the two sides? There is reason to believe that in this case the tunneling supercurrents are nondissipative and of essentially the same nature as ordinary supercurrents. For example, in a closed ring containing a barrier there is a set of different solutions to the modified Ginzburg-Landau equations, in which the phase of ψ changes by different multiples of 2π in going round the ring. Jumping from one solution to the other involves overcoming the coupling energy, and if this is impossible the higher energy solutions, which are current carrying ones, will be metastable.

Magnetic fields can pass between two superconductors separated by a barrier, in the form of a thin sheet of flux penetrating a short distance into the superconductors in the usual way. In the presence of flux ϕ varies over the barrier in a way determined by the following rule.[10] Consider two points P and Q on opposite sides of the barrier (and flux sheet) joined by two curves crossing the barrier at different points A and B. The difference between the phases of ϕ at A and B is proportional to the flux between the two curves, one flux quantum *hc/2e* corresponding to a phase difference of 2π. This effect is closely related to the one predicted by Ehrenberg and Siday[11] for interference between electron beams in the presence of a magnetic field.

In differential form we may write

$$\text{grad } \phi = \frac{2ed}{\hbar c} (\mathbf{H} \times \mathbf{n}) . \tag{2.1}$$

Here **H** is the field in the barrier, **n** is a unit vector normal to the barrier, and d is the effective thickness of the sheet of flux:

$$d = \lambda_1 + \lambda_2 + t , \tag{2.2}$$

where λ_1, λ_2 are the penetration depths on the two sides, and t is the barrier thickness.[12]

Combining (2.1) and (1.3) and Maxwell's equations, we obtain [10,13]

$$\nabla^2 \phi = (\lambda')^{-2} \sin \phi , \tag{2.3}$$

where

$$\lambda' = (\hbar c^2 / 8\pi e j_1 d)^{\frac{1}{2}} . \tag{2.4}$$

Here ∇^2 is a twodimensional operator. λ' is typically of the order of 1 mm. and has the significance of a penetration depth, as can be seen by putting $\sin \phi \sim \phi$ in (2.3).

The detailed behavior of coupled superconductors depends on the transverse dimensions of the barrier

5 Sign conventions: if the sides of the barrier are labeled 1 and 2 and n is a unit vector pointing from 1 to 2 then $\phi = \arg \psi_2 - \arg \psi_1$, $j = \mathbf{j} \cdot \mathbf{n}$, $V = V_1 - V_2$.

6 The proof is similar to that used to find the energy stored in an inductor, and is based on Eq. (3.2).

7 P. W. Anderson, *Proceedings of the Ravello Spring School,* 1963 (to be published).

8 P. W. Anderson and J. M. Rowell, Phys. Rev. Letters 10, 230 (1963).

9 This will be discussed in more detail in Sec. IV.

10 P. W. Anderson (private communication).

11 W. Ehrenberg and R. E. Siday, Proc. Phys. Soc. (London) B62, 8 (1949). In the present case the flux quantum is half that in the case of interference of electron beams, since electron pairs are involved (Sec. III).

12 Note that d is not simply the barrier thickness as has sometimes been supposed.

13 R. A. Ferrell and R. E. Prange, Phys. Rev. Letters 10, 479 (1963).

relative to λ'. Barriers large compared with λ' behave in a similar manner to superconductors of the second kind.[14] In weak magnetic fields diamagnetic currents screen out the field from the barrier except at points within a distance of the order of λ' from the edge. At a critical field[15]

$$H_c' = (32\hbar j_1/\pi e d)^{\frac{1}{2}} \tag{2.5}$$

of the order of 1 G, a second order phase transition occurs and quantized flux lines start to penetrate into the barrier, their separation decreasing as the field is increased. This is the ideal thermodynamic behavior; in practice it would appear that hysteresis may occur, owing to the attachment of flux lines to the edge of the barrier.[15] In the case considered by Ferrell and Prange,[13] a field of $\frac{1}{2}\pi$ times the critical field H_c' is required before flux starts penetrating. Owing to screening, when a supercurrent is passed through a barrier it is normally all carried in the neighborhood of the edges.[10] Ferrell and Prange[13] have shown that the effective current-carrying width of a barrier is $2\lambda'$. However, if a barrier is inhomogeneous, it may be possible for current to be carried by its interior, owing to flux pinning, just as in type II superconductors.[16,17]

The behavior of a barrier small compared with λ' is somewhat simpler. Just as with very thin fihns, the magnetic field inside such a barrier is almost constant. The main feature of interest is the field dependence of the critical supercurrent. As we have seen, the effect of a field is to cause ϕ to vary in phase over the barrier. Therefore, in view of (1.3), in a sufficiently strong field the barrier is split up into regions in which the current through it has opposite signs, and the maximum value of the total supercurrent is thus greatly reduced. The field dependence of the critical current is given by a Fraunhofer diffraction pattern formula for an aperture of the same shape as the barrier (and transmission amplitude j_1).[18] For a rectangular barrier with the field along a side, one obtains

$$I_c \propto \left|\frac{\sin\,(H/H_0)}{H/H_0}\right|, \tag{2.6}$$

where $H_0/2\pi$ is the field for which the barrier encloses one flux quantum $hc/2e$. Behavior of this type has been observed by Rowell,[18] and this experiment, together with the previous one of Anderson and Rowell,[8] indicates strongly that the currents observed are indeed tunneling through the barrier and not going through isolated metallic bridges.

III. NONEQUILIBRIUM PROPERTIES

In dealing with the nonequilibrium properties of coupled superconductors, the first requirement is to find a formula for the time dependence of ϕ. Here I give a not very rigorous derivation, which serves, however, to introduce the important topic of the interaction of tunneling supercurrents with photons.[19]

So far we have-been using what may be called the wave picture of a superconductor-we have been dealing with ψ, the wave function of the superconducting electrons. From the microscopic theory of superconductivity it is known that the particles associated with the wave function are bound pairs of electrons called Cooper pairs.[4,20] In some ways a superconductor can be thought of as a Bose-Einstein condensation of Cooper pairs. The supercurrents we have been discussing may be viewed as Cooper pairs tunneling through the barrier [Fig. l(a)]. If there is a

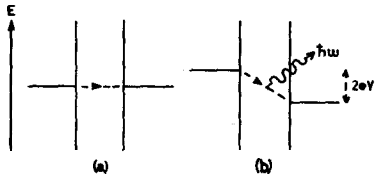


FIG. 1. Quasi-particle picture of tunneling supercurrents. (a) dc supercurrents at zero voltage. (b) ac suppercurrents at nonaero voltage, giving rise to the emission of photons. Corresponding to each emitted photon a Cooper pair tunnels across the barrier in the direction of the applied voltage, conservation of energy requiring that $\hbar\omega = 2eV$. In the presence of a microwave field this process and its reverse can occur, giving rise to photon-induced supercurrents.

nonzero potential difference V between the two sides of the barrier, Cooper pairs on different sides of the barrier have energies differing by $\Delta E = 2eV$ (the charge of a pair being $2e$). Tunneling through the barrier can then take place only as a virtual process. We know from elementary quantum mechanics that in such a system there are oscillating currents at a frequency $\nu = \Delta E/h$, i.e.,

$$\nu = 2eV/h. \tag{3.1}$$

When the interaction between the oscillating currents and the electromagnetic field is taken into account, one finds that real processes can take place,

[14] A. A. Abrikosov, Zh. Eksperim. i Teor. Fis. 32, 1442 (1957) [English transl.: Soviet Phys.—JETP 5, 1174 (1957)].
[15] B. D. Josephson (unpublished work).
[16] P. W. Anderson, Phys. Rev. Letters 9, 309 (1962).
[17] For example, a superconducting bridge across the barrier would have this effect, and a region with a high local value off would behave similarly.
[18] J. M. Rowell, Phys. Rev. Letters 11, 200 (1963).
[19] A more rigorous proof is given in reference 1. An alternative proof based on gauge invariance has been given by Anderson (reference 7).
[20] L. N. Cooper, Phys. Rev. 104, 1189 (1956).

energy being conserved by the emission of a photon [Fig. l(b)]. In this case the radiation is coherent, since every photon comes from an identical process.

Taking into account the fact that the oscillating currents arise from changes in ϕ, we see that

$$\dot{\phi} = 2\pi\nu = 2eV/\hbar . \tag{3.2}$$

Now let us consider what happens if we apply to the barrier radiation of the same frequency as the oscillating supercurrents. When two oscillating systems are coupled together, energy can be transferred in either direction, depending on the phase relationships. The energy transfer process in this case is Fig. 1(b), or the reverse process.[21] Therefore, in the presence of microwave radiation dc supercurrents (i.e. the transfer of Cooper pairs across the barrier) can occur provided the potential difference is such that energy can be conserved by absorption or stimulated emission of a photon (multiphoton processes can also occur). This process is characterized by the appearance of constant voltage regions in the *I-V* characteristic, and this has recently been observed by Shapiro,[22] who made the interesting observation that for particular values of microwave power the specimen would spontaneously jump on to such a constant voltage region. Shapiro's specimen is therefore behaving as, an ideal zero-impedance voltage source, powered by the microwave field.

For the supercurrent-photon coupling to be strong it is necessary that both the supercurrents and the electric field in the barrier produced by the microwave radiation should be in phase all over the barrier. Hence small barriers are favorable to the observation of the effect.

I shall now pass on to a different type of nonequilibrium effect. This concerns what happens when one applies rf fields to a system in which both sides of the barrier are originally at the same potential. If the rf fields are small enough they do not break up phase coherence across the barrier but merely cause oscillations in phase. As may be seen by taking the time derivative of (1.3) and using (3.2), the barrier behaves with respect to supercurrents exactly like an inductance $L_0 \sec\phi$ per unit area, where

$$L_0 = \hbar/2ej_1 . \tag{3.3}$$

One must also take into account the capacitance C per unit area (as indeed one must to treat correctly the earlier work in this section). The following dispersion equation is obtained,[15] assuming dc fields and currents to be absent.[23]

$$(\omega/\omega_0)^2 = (k\lambda')^2 + 1 , \tag{3.4}$$

where

$$\omega_0 = (L_0C)^{-\frac{1}{2}}, \tag{3.5}$$

which corresponds to frequencies of a few Gc/sec. When $\omega > \omega_0$, traveling waves localized in the neighborhood of the barrier can be propagated along it. These could be excited by applying alternating voltages across the barrier at its edge. When $\omega < \omega_0$, k is imaginary, and disturbances excited in this manner decay exponentially with distance from the edge, just as in the dc limit considered in Sec. II. The impedance across unit length of the edge of a semi-infinite barrier is $4\pi d\omega/kc^2$ (inductive when k is imaginary),[15] which is normally very small ($< 10^{-3}$ Ω cm) except when $\omega \sim \omega_0$, when the impedance becomes theoretically infinite. When $\omega = \omega_0$, $k = 0$ and the disturbance is a type of plasma oscillation, with the current and electric field normal to the barrier and the magnetic field absent. The power that could be transmitted along a barrier without breaking up phase coherence should be sufficient to permit direct demonstration of the existence of the cutoff frequency ω_0.

A final point of interest in connection with this type of effect concerns quasi-particle currents. According to the microscopic theory,[1] the barrier conductivity which determines these is phase dependent:

$$\sigma = \sigma_0 + \sigma_1 \cos\phi . \tag{3.6}$$

The phase dependent term does not affect the dc *I-V* characteristics of the barrier because cos ϕ averages out to zero, but it can affect the quasi-particle damping of waves of the type just considered.

IV. THERMAL AND ZERO-POINT FLUCTUATIONS

For very small barriers the only mode of oscillation important in producing fluctuations in ϕ is the $k = 0$, $\omega = \omega_0$ mode of Sec. III, in which the oscillations in ϕ are in phase all over the barrier. Assuming fluctuations in ϕ are small the oscillations can be assumed to he simple harmonic, and we obtain for the mean square deviation of ϕ:

$$(\Delta\phi)^2 = \frac{2e\omega_0}{j_1 A}\left[\frac{1}{2} + \frac{1}{\exp(\hbar\omega_0/kT) - 1}\right], \tag{4.1}$$

where A is the area of the barrier. As the barrier is made larger, however, modes with $k \neq 0$ become important and eventually $(\Delta\phi)^2$ must become inde-

[21] Similar processes for tunneling quasi-particles have been observed by A. H. Dayem, and R. J. Martin [Phys. Rev. Letters 8, 246 (1962)] and discussed theoretically by P. K. Tien and J. P. Gordon [Phys. Rev. 129, 647 (1963)].

[22] S. Shapiro, Phys. Rev. Letters 11, 80 (1963).

[23] In general both j_1 and λ' may be frequency dependent, the latter through the frequency dependence of λ.

pendent of A. Clearly in order to observe effects which depend on having phase coherence over the barrier, one must have $\overline{(\Delta\phi)^2} \ll 1$. This sets a lower limit to the value of j_1. As pointed out by Anderson and Rowell,[8] the temperature seen by the barrier is not necessarily the cryostat temperature, since thermal noise is transmitted to the specimen down the leads to the cryostat from room. temperature circuitry. Since this is at a temperature of the order of a hundred times greater, it is clearly desirable to choose conditions so that this effect is minimized. To see the conditions necessary for this, consider the idealized circuit of Fig. 2. The specimen is represented by a damped tuned circuit and the room temperature source by a noise voltage V_N in series with a resistor R_1. It is seen that the object of reducing the current through L can be achieved by

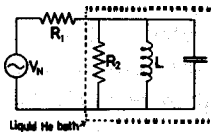


FIG. 2. Simplified circuit diagram illustrating the effects of electrical noise transmitted down the leads to the cryostat.

(i) increasing the source impedance R_1;

(ii) inserting an electrical filter cooled to liquid helium temperature in the leads to the specimen. In particular, frequencies of ω_0 or above where there may be high Q resonances should be removed;

(iii) if possible, damping out the resonances by reducing R_2, i.e., connecting a normal metal shunt across the barrier.

V. MICROSCOPIC THEORY

Various methods have been used to calculate j_1 from basic theory.[1,7,24,25] Many of these use the tunneling Hamiltonian method of Cohen *et al.*[26] The simplest method is that of Anderson,[7] who uses statistical mechanical perturbation theory to calculate f and then derives j_1 from (1.2). Formulas for j_1 have been given for absolute zero by Anderson[7] and calculated numerically for finite temperatures by Ambegaokar and Baratoff.[24] This method cannot be used for nonequilibrium processes, which can be dealt with by adiabatic perturbation theory.[1] An important result of the microscopic theory is that j_1 is proportional to the tunneling probability for quasi-particles, so that tunneling supercurrents are similar in magnitude to quasi-particle currents at voltages of the order of the energy gap.

Objections have been raised against the tunneling Hamiltonian method on the grounds that it treats the barrier as a mathematical plane and ignores events taking place in it.[27] A method which avoids this difficulty is to describe propagation through the barrier by Green's functions.[15,25] This method has the additional advantages that it shows precisely the conditions necessary for coupling to occur, it indicates the relation between supercurrents through barriers and the usual kind of supercurrent, and it is not restricted to insulating barriers. Gor'kov's theory[4] allows one to express the current density in a superconductor as an expression in powers of ψ and ψ^*. The lowest order term has the form

$$\mathbf{j}(\mathbf{r}) = \int \mathbf{K}(\mathbf{r},\mathbf{r}',\mathbf{r}'')\psi^*(\mathbf{r}')\psi(\mathbf{r}'')d\mathbf{r}'d\mathbf{r}''. \qquad (5.1)$$

If ψ is assumed to have a linear variation near $\mathbf{r}$, this reduces to the Ginzburg-Landau equations. A more appropriate approximation for barriers is to assume that ψ is slowly varying on each side of the barrier but changes suddenly as one crosses the barrier; this assumption leads to Eq. (1.3).

An important feature of (5.1) is that the relation between $\mathbf{j}$ and ψ is nonlocal. This leads to qualitatively different results from those of Bardeen,[28] who assumed a local relation. Detailed consideration of (5.1) shows that coupling is not dependent on an interaction in the barrier, but on the ability of electrons to propagate elastically through the barrier (as described by the single-electron Green's function). It is further necessary that time-reversal symmetry should apply to the propagation process. Finally, from (5.1) one can deduce that repulsive interactions in the barrier are, as one might expect, detrimental to coupling.

VI. CONCLUSION

With the exception of the last section, my aim in this paper has been to show that just as in the days before the BCS theory one knew a great deal about the behavior of superconductors purely on the basis of phenomenological theories, one can similarly predict many varied types of behavior of barriers between superconductors on the basis of a few simple equations based on straightforward assumptions. These equations can be justified by the microscopic theory, and, as far as present experimental evidence goes, seem to give a substantially correct description of events.

[24] V. Ambegaokar and A. Baratoff, Phys. Rev. Letters 10, 486 (1963) and 11, 104 (1963) (erratum).
[25] P. G. de Gennes, Phys. Letters 5, 22 (1963).
[26] M. H. Cohen, L. M. Falicov, and J. C. Phillips, Phys. Rev. Letters 8, 316 (1962).
[27] J. Bardeen (private communication).
[28] J. Bardeen, Phys. Rev. Letters 6, 57 (1961) and 9, 147 (1962).

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